



جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم الهندسة الكهربائية
مختبر الكترونيات 1

2023

❖ **Vision:**

To produce professionally competent Electronics and Communication Engineers capable of effectively and efficiently addressing the technical challenges with social responsibility.

❖ **Mission:**

The mission of the Department is to provide an academic environment that will ensure high quality education, training and research by keeping the students abreast of latest developments in the field of Electronics and Communication Engineering aimed at promoting employability, leadership qualities with humanity, ethics, research aptitude and team spirit.

❖ **Quality Policy:**

Our policy is to nurture and build diligent and dedicated community of engineers providing a professional and unprejudiced environment, thus justifying the purpose of teaching and satisfying the stake holders.

A team of well qualified and experienced professionals ensure quality education with its practical application in all areas of the Institute.

❖ **Philosophy:**

The essence of learning lies in pursuing the truth that liberates one from the darkness of ignorance and Institute of Aeronautical Engineering firmly believes that education is for liberation.

Contained therein is the notion that engineering education includes all fields of science that plays a pivotal role in the development of world-wide community contributing to the progress of civilization. This institute, adhering to the above understanding, is committed to the development of science and technology in congruence with the natural environs. It lays great emphasis on intensive research and education that blends professional skills and high moral standards with a sense of individuality and humanity. We thus promote ties with local communities and encourage transnational interactions in order to be socially accountable. This accelerates the process of transfiguring the students into complete human beings making the learning process relevant to life, instilling in them a sense of courtesy and responsibility.

Program Outcomes	
PO1	Engineering knowledge: An ability to apply knowledge of basic sciences, mathematical skills, engineering and technology to solve complex electronics and communication engineering problems (Fundamental Engineering Analysis Skills).
PO2	Problem analysis: An ability to identify, formulate and analyze engineering problems using knowledge of Basic Mathematics and Engineering Sciences. (Engineering Problem Solving Skills).
PO3	Design/development of solutions: An ability to provide solution and to design Electronics and Communication Systems as per social needs (Social Awareness)
PO4	Conduct investigations of complex problems: An ability to investigate the problems in Electronics and Communication field and develop suitable solutions (Creative Skills).
PO5	Modern tool usage An ability to use latest hardware and software tools to solve complex engineering problems (Software and Hardware Interface).
PO6	The engineer and society: An ability to apply knowledge of contemporary issues like health, Safety and legal which influences engineering design (Social Awareness).
PO7	Environment and sustainability An ability to have awareness on society and environment for sustainable solutions to Electronics & Communication Engineering problems (Social awareness).

PO8	Ethics: An ability to demonstrate understanding of professional and ethical responsibilities (Engineering impact assessment skills).
PO9	Individual and team work: An ability to work efficiently as an individual and in multidisciplinary teams (Team Work).
PO10	Communication: An ability to communicate effectively and efficiently both in verbal and written form (Communication Skills).
PO11	Project management and finance: An ability to develop confidence to pursue higher education and for life-long learning (Continuing education awareness).
PO12	Life-long learning: An ability to design, implement and manage the electronic projects for real world applications with optimum financial resources (Practical engineering analysis skills).
Program Specific Outcomes	
PSO1	Professional Skills: An ability to understand the basic concepts in Electronics & Communication Engineering and to apply them to various areas, like Electronics, Communications, Signal processing, VLSI, Embedded systems etc., in the design and implementation of complex systems.
PSO2	Problem-solving skills: An ability to solve complex Electronics and communication engineering problems, using latest hardware and software tools, along with analytical skills to arrive cost effective and appropriate solutions.
PSO3	Successful Career and Entrepreneurship: An understanding of social-awareness & environmental-wisdom along with ethical responsibility to have a successful career and to sustain passion and zeal for real-world applications using optimal resources as an Entrepreneur.

❖ **Lab Course Overview:**

This Lab course provides the basic knowledge over the construction and functionality of the basic electronic devices such as diodes and transistors. It also provides the information about the uncontrollable and controllable electronic switches and the flow of current through these switches in different biasing conditions. This Lab course is intended to describe the different configurations and modes of **controllable** switches and how these electronic devices can be configured to work as rectifiers, clippers, clampers, Zener and Transistors Characteristics (BJT & FET).

❖ **Course Outcomes:**

Upon the completion of Data Structures practical course, the student will be able to:

1. Analyze the diode and transistor characteristics.
2. Understand the principles of rectifier circuits using diodes and implement them using hardware.
3. Design the biasing circuits like self-biasing.
4. Understand the concepts of BJT and observe its characteristics.
5. Understand the concepts of unipolar junction transistor (FET) and observe its characteristics.

❖ **Instructions to the Students:**

1. Students are required to attend all labs.
2. Students work individually and in groups in the hardware and software laboratories.
3. Students have to bring the pre-lab manual along with them whenever they come for lab work.
4. Should take only the lab manual, calculator (if needed) and a pen or pencil to the work

area.

5. Should learn the prelab questions. Read through the lab experiment to familiarize themselves with the components and assembly sequence.
6. Should utilize 3 hours time properly to perform the experiment and to record the readings. Do the calculations, draw the graphs and take signature from the instructor.
7. If the experiment is not completed in the stipulated time, the pending work has to be carried out in the leisure hours or extended hours.
8. Should submit the completed record book according to the deadlines set up by the instructor.
9. For practical subjects there shall be a continuous evaluation during the semester consist of:
 - A. 20 sessional marks (*Lab reports*).
 - B. 30 marks (*mid-term exam*).
 - C. 50 marks (*final examination*).

❖ **The Electric Shock:**

The electric Shock is a cause of an electric current passing through the human body, and the severity depends mainly on its amount. The main source of electric shock is improper equipment grounding. Nowadays most of equipment are produced with a three wire cord and thus they are safer to use. The third wire that is connected to a metal case is also connected to the earth ground (usually a pipe or bar in the ground) through the wall plug outlet.

❖ **Basic Safety Precautions:**

All students and users are advised to comprehend to the following safety precautions whenever they work in the laboratories:

1. Acquaint yourself with the location of the following safety items (***Fire Extinguisher, First aid kit, Circuit Breaker, and emergency Telephone numbers***).
2. Make sure that the lab instruments are at ground potential by using the ground terminal supplied on the instrument. Never handle wet, damped, or ungrounded electrical equipment.
3. Never touch electrical equipment while standing on a damp or metal floor.
4. In an emergency all power in the laboratory can be switched off by depressing the large red button on the main breaker panel.
5. Do not wear bracelets, rings and watches during lab sessions.
6. Power must be switched off whenever an experiment or project is being assembled, disassembled, or modified. Discharge any high voltage points to grounds with a well-insulated jumper. Remember that capacitors can store dangerous quantities of energy.
7. Make measurements with well insulated probes, and avoid using both hands.
8. If a person comes in contact with a high voltage, immediately shut off the power. Do not attempt to remove a person in contact with a high voltage unless you are insulated from him/her. If the victim is not breathing, apply CPR immediately continuing until he/she is revived, and have someone dial emergency numbers for assistance.
9. Use CO₂ to put off an electrical fire or a dry type fire extinguisher. Locate extinguishers and read operating instructions before an emergency occurs.

❖ **Electronics Workshop Practice:**

Identification, specifications, testing of R, L, C components (Color Codes), potentiometers, switches (SPDT, DPDT and DIP), coils, Gang condensers, relays, bread boards, PCBs, identification, specifications and testing of active devices, diodes, BJTs, Low power JFETs, MOSFETs and Op-Amp.

▪ **Equipment and Materials:**

A. Equipment:

1. DC Power Supply:

A Dual DC power supply figure 1 can be used (1) as a voltage Source, and (2) as a current Source provided a prior setting of the unit. The DC power supply provided in the laboratory is set to operate as voltage source, and the controls are:

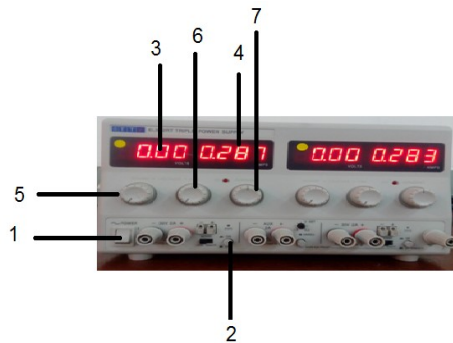


Figure 1: Digital Dual DC Power Supply

Control No.	Description
1	Main power Switch: Provides power to the unit
2	Secondary Switch: Provides power to the circuit.
3 & 4	Voltage and Current monitors
5	Coarse: Enables voltages from (0 to 30)V range.
6	Fine: Enables tuning the voltage value.
7	Current Limit (CC): Should be at Mid-point to enable each source to deliver a sufficient current to the Load.

2. Function Generator:

A function generator figure (2) is a device that can produce various patterns of voltage at a variety of frequencies and amplitudes.



(A)

(B)

(C)

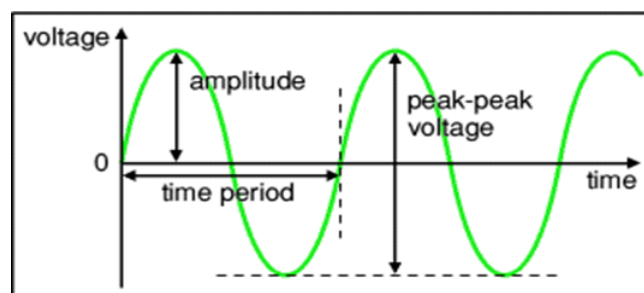
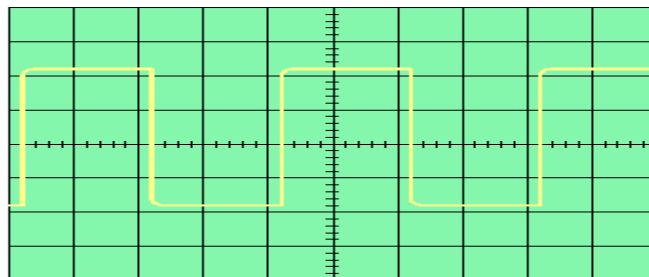
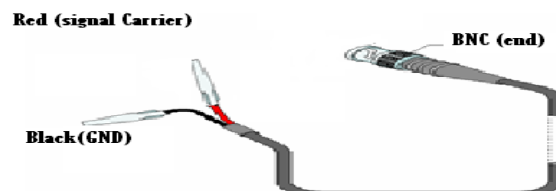
The Oscilloscope:

The oscilloscope, either digital or analog, is the most useful instrument available for testing circuits because it allows you to see the signals at different points in the circuit. The place or type of control buttons found on any oscilloscope front panel may differ according to the manufacturer's preference see figure (3), but the functions are the same. It draws a V/t graph (voltage verses time), voltage on the vertical or (Y-axis), and time on the horizontal or (X-axis).



BNC Probe:

It is a special connector that has two Ends. BNC is connected into the oscilloscope and two clip jacks that represent signal carrier and GND. The channel and the probe can be checked by the BNC to the input of CH1 or CH2 and adjusting V/DIV to 2V/DIV. A square wave similar to that of figure 4 will be shown on the Oscilloscope screen.



3. **Multimeters:**

Aim: To accurately read voltages using analog voltmeter and DMM from Regulated power supply

Theory: Analog Voltmeter:

Although digital meters are used extensively throughout the electronic industry. There are many analog meters also in use. The DC voltage scales on analog voltmeter are linear that is the distance between equal values (adjacent divisions) marked on the meter scale are of same length.

Ex: If the scale is from 0V to 10V then there are 10 equally spaced divisions between these two values each division represents 0.2V.

Zero adjustment:-

Before use make any measurement with the voltmeter, be certain that the meter indication 0V. In analog voltmeter, control (screw) is placed on the front panel of the meter for the purpose of zeroing the meter.

Parallax Error:

It results when the person making the measurement is not directly viewing the meter pointer.

Theory: Digital Multimeter:

A Multimeter is an electronic device that is used to make various electrical measurements, such as AC and DC voltage, AC and DC current, and resistance. It is called a Multimeter because it combines the functions of a voltmeter, ammeter, and ohmmeter. Multimeter may also have other functions, such as diode test, continuity test, transistor test, TTL logic test and frequency test.

Parts of Multimeter:

A Multimeter has three parts:

- Display
- Selection Knob
- Ports

The display usually has four digits and the ability to display a negative sign. A few multimeters have illuminated displays for better viewing in low light situations.

The selection knob allows the user to set the multimeter to read different things such as milliamps (mA) of current, voltage (V) and resistance (Ω).

Two probes are plugged into two of the ports on the front of the unit. COM stands for common and is almost always connected to Ground or „-“ of a circuit. The COM probe is conventionally black but there is no difference between the red probe and black probe other than color. 10A is the special port used when measuring large currents (greater than 200mA). mA/V Ω is the port that the red probe is conventionally plugged in to. This port allows the measurement of current (up to 200mA), voltage (V), and



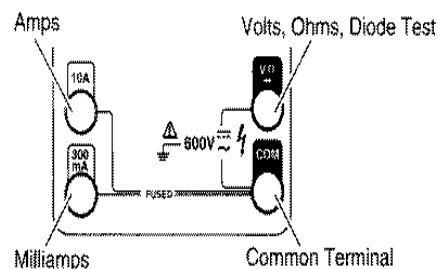
resistance (Ω). The probes have a banana type connector on the end that plugs into the multimeter. Any probe with a banana plug will work with this meter.

Safety Measures:

- Be sure the test leads and rotary switch are in the correct position for the desired measurement.
- Never use the meter if the meter or the test leads look damaged.
- Never measure resistance in a circuit when power is applied.
- Never touch the probes to a voltage source when a test lead is plugged into the 10 A or 300 mA input jack.
- To avoid damage or injury, never use the meter on circuits that exceed 4800 watts.
- Never apply more than the rated voltage between any input jack and earth ground.
- Be careful when working with voltages above 60 V DC or 30 V AC rms. Such voltages pose a shock hazard.
- Keep your fingers behind the finger guards on the test probes when making measurements.
- To avoid false readings, which could lead to possible electric shock or personal injury, replace the battery as soon as the battery indicator appears.

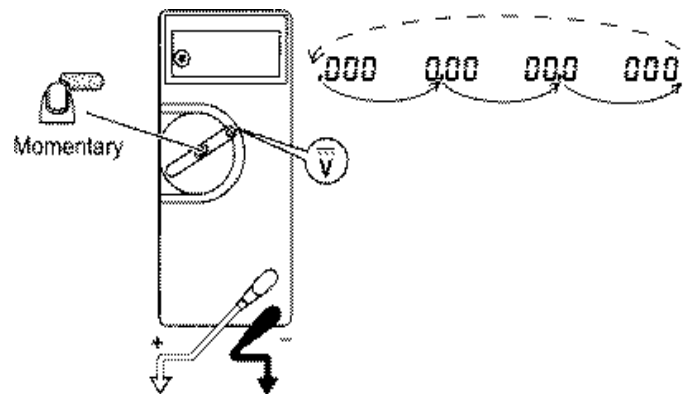
Input Jacks:

The black lead is always plugged into the common terminal. The red lead is plugged into the 10 A jack when measuring currents greater than 300 mA, the 300 mA jack when measuring currents less than 300 mA, and the remaining jack (V-ohms-diode) for all other measurements.



Range Fixing:

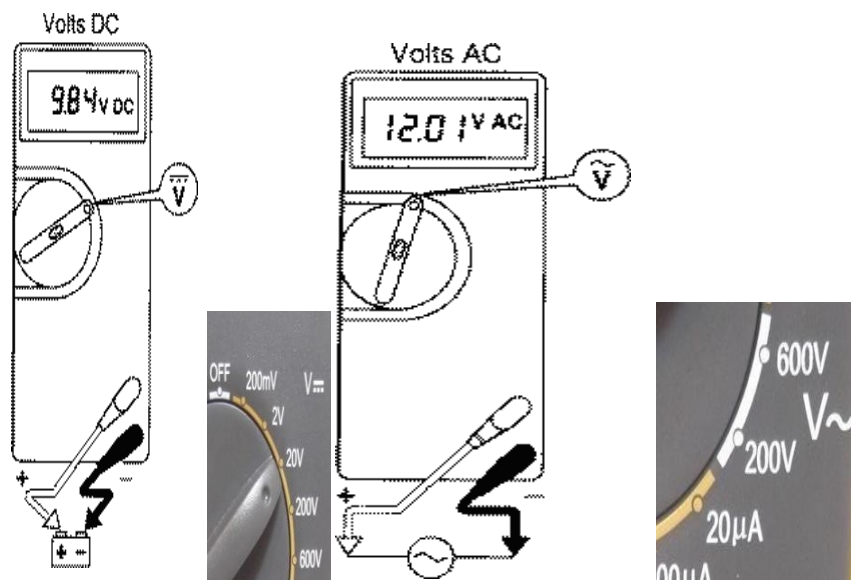
The meter defaults to auto range when first turned on. You can choose a manual range in V AC, V DC, A AC, and A DC by pressing the button in the middle of the rotary dial. To return to auto-range, press the button for one second.



■ Procedure for Measurement Voltage Measurement

A.C. Voltage Measurement

- Connect the positive (red) test lead to the “V/mA” jack socket and the negative (black) lead to the “COM” jack socket.
- Set the selector switch to the desired mV D.C./D.C.V/A.C.V range.
- Connect the test leads to the circuit to be measured.
- Turn on the power to the circuit to be measured, the voltage value should appear on the digital display along with the voltage polarity (if reversed only).

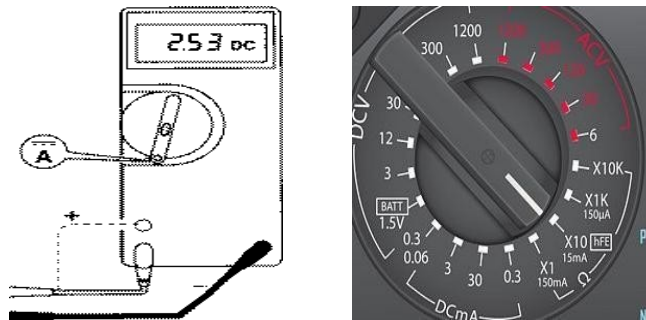


Current Measurement

1. Connect the positive (red) test lead to the “V/mA” jack socket and the negative (black) lead to the “COM” jack socket (for measurements up to 200mA). For measurements between 200mA and 10A connect the red test lead to the “10mA” socket.
2. Set the selector switch to the desired μA /mA/A range.
3. Open the circuit to be measured and connect the test leads in SERIES with the load

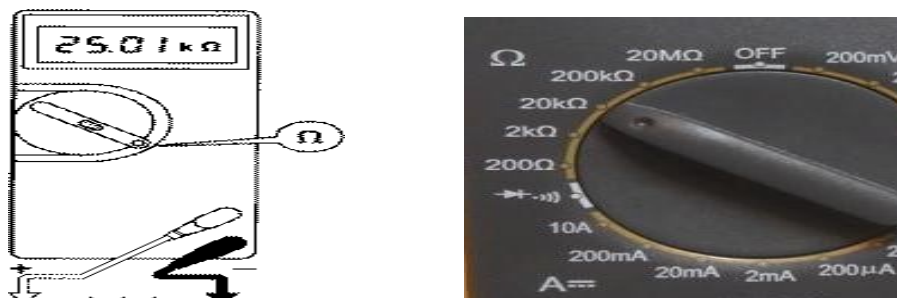
in which current is to be measured.

4. To avoid blowing an input fuse, use the 10A jack until you are sure that the current is less than 300 mA. Turn off power to the circuit. Break the circuit. (For circuits of more than 10 amps, use a current clamp). Put the meter in series with the circuit and turn power on.



Resistance Measurement

1. Connect the positive (red) test lead to the “V/mA” jack socket and the negative (black) lead to the “COM” jack socket.
2. Set the selector switch to the desired “OHM Ω ”.
3. If the resistance to be measured via part of a circuit, turn off the power and discharge all capacitors before measurement.
4. Connect the test leads to the circuit to be measured.
5. The resistance value should now appear on the digital display.
6. If the resistance to be measured is part of a circuit, turn off the power and discharge all capacitors before measurement.

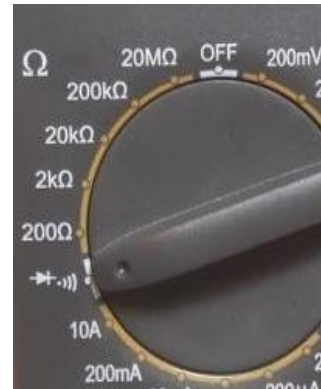
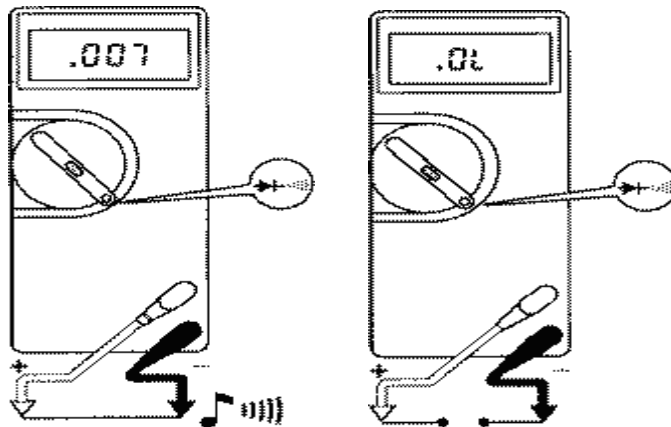


Continuity Test

This mode is used to check if two points are electrically connected. It is often used to verify connectors. If continuity exists (resistance less than 210 ohms), the beeper sounds continuously.

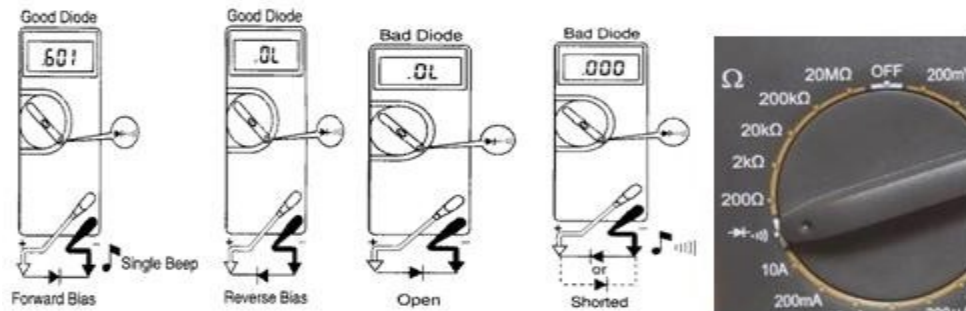
1. Connect the positive (red) test lead to the “V/mA” jack socket and the negative (black) lead to the “COM” jack socket.
2. Set the selector switch to the  position.
3. Connect the test leads to two points of the circuit to be tested. If the resistance is Ohms the buzzer will sound.

4. If the resistance to be measured is part of a circuit, turn off the power and discharge all capacitors before measurement.



Diode Test

1. Connect the positive (red) test lead to the “V/mA” jack socket and the negative (black) lead to the “COM” jack socket.
2. Set the selector switch to the position.
3. Connect the test leads to be measured.
4. Turn on the power to the circuit to be measured and the voltage value should appear on the digital display.



General Operation: Connection of Probes:

All multimeters come with two probes. They are to be connected to the terminals on the meter itself. The Black probe is to be connected to the COM terminal. Red probe is to be connected to terminal marked with:

- V- Ω for voltage measurement.
- “mA” or “20A” for current measurement (there are two terminals, one for 2A range and the other for 20A range)
- V- Ω for resistance measurement.

Setting of function:

The multimeter uses different circuits internally to measure different things. Therefore, you must select the correct function before using it.

Setting of Range:

- You can change the sensitivity of the meter by selecting different range for measurement. Set the range to the first range that is higher than the maximum value you expect to measure. This will give a more accurate reading. If you do not know what to expect, use the highest range first.
- After a reading is obtained, set the range to the appropriate one to get a better reading.
- When the value measured exceeds the existing range, the display will flash. When this happens, set the multimeter to a higher range until some values are displayed.

Precautions:

1. For current measurement, the maximum input current is 2A (if the RED mA and BLACK COM terminals are used) or 20A (if the RED 20A and BLACK COM terminals are used). Excessive current will blow the fuse on the 2A range, which must be replaced. The 20A range, however, is not protected by the fuse.
2. For voltage measurement (use the RED V- Ω and BLACK COM terminals), the maximum input voltage on all DC ranges is 1200V DC or peak AC
 - the 20V, 200V and 1000V AC ranges is 1000Vrms continuous

- the 2V and 200mV AC ranges is 1000Vrms for not more than 15 seconds.
3. To avoid electrical shock and/or instrument damage, do not connect the COM input terminals to any source of more than 500 volts DC or peak AC above earth ground.

Resistance measurement

Procedure:

- Connect probes: black probe to COM terminal and red probe to terminal marked with “Ω”
- Set function to resistance measurement
- Set to the appropriate range (refer to above)
- Connect the two probes” crocodile clips to the resistor (or to the resistor circuit via jumper wires) to make measurement
- Note the reading, adjust range if necessary
- Take the more accurate reading.

Voltage measurement:

Procedure:

- Connect probes: black probe to COM terminal and red probe to terminal marked with “V”
- Set function to voltage measurement
- Set to the appropriate range (refer to above)
- Set the AC-DC selection - depends on what type of signal you want to measure
- Touch the two points where you want to make measurement
- Note the reading, adjust range if necessary
- Take the more accurate reading.

Note:

Reading obtained is the voltage of where the red probe touch with reference to where the black probe touches. This may not indicate the voltage level from ground. To find the voltage level of a point from ground, black probe should be touching a ground point and the red probe on the point you want to measure.

Current measurement:

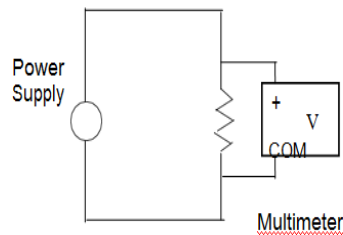
Procedure:

- Connect probes: black probe to COM terminal and red probe to terminal marked with “A”
- Set function to current measurement
- Set to the appropriate range
- Set AC-DC selection - depends on what type of signal you want to measure
- Off the power to the circuit
- Break the path which we want to make measurement
- Connect the path with the two probes so that current now flow through the multimeter on the power
- Note the reading, change range if necessary
- Take the more accurate reading.

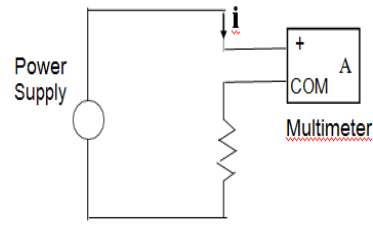
Note

Use the 20A range if you are not sure of the current to be measured.

Connection Methods to Measure Voltage and Current



Voltage Measurement



Current Measurement

5. The Breadboard:

It is an experimental board that has many strips of metal (usually copper) which run underneath as in figure (A). The strips are covered by a ceramic or a plastic surface that has holes that match the strips of metal and the finished breadboard is as shown in figure (B). Figure (B) also shows how the components should be placed on the breadboard.

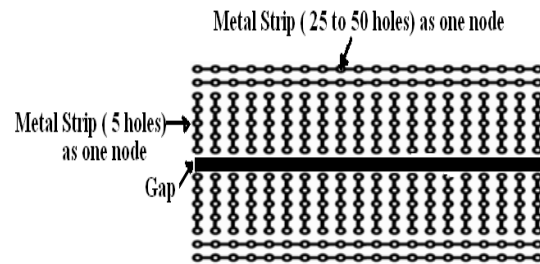
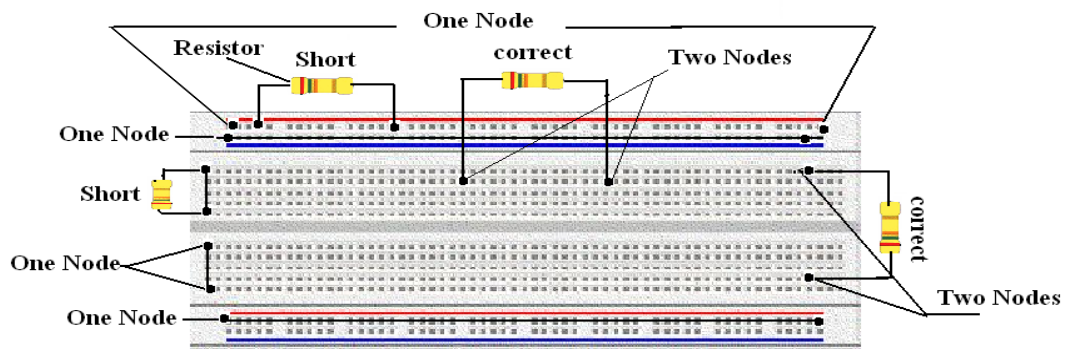


Figure A : connection layout diagram



B. Materials:

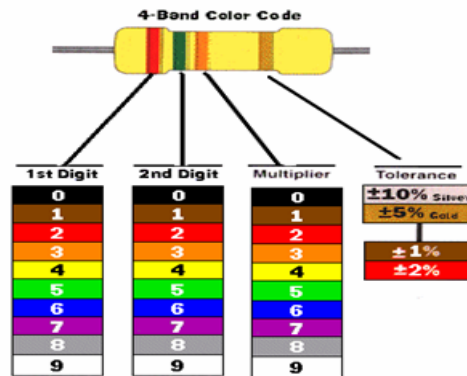
1. RESISTOR:

A resistor value is determined according to a standard color code as shown in figure below. In practice however; every resistor has a tolerance that determines its validity; the tolerance is determined by gold, silver or no color band.

The standard resistor color code table:

Color	Digit 1	Digit 2	Multiplier	Tolerance
Black	0	0	$\times 10^0$	
Brown	1	1	$\times 10^1$	$\pm 1\%$
Red	2	2	$\times 10^2$	$\pm 2\%$
Orange	3	3	$\times 10^3$	
Yellow	4	4	$\times 10^4$	
Green	5	5	$\times 10^5$	
Blue	6	6	$\times 10^6$	
Violet	7	7	$\times 10^7$	
Gray	8	8	$\times 10^8$	
White	9	9	$\times 10^9$	
Gold			$\times 0.1$	$\pm 5\%$
Silver			$\times 0.01$	$\pm 10\%$
None				$\pm 20\%$

A: color Code



B: Resistor Bands

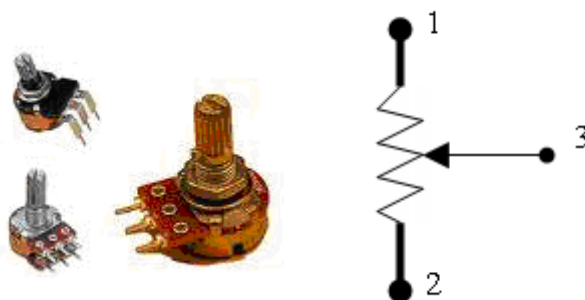
Resistor identification (A): Color Code, (B): Resistor Bands

Applications:

It is widely used in electronic circuits to limit the current

2. Potentiometer:

It is a variable resistor that has three terminals as shown in figure below. It could be used as a fixed resistor when terminals (1 and 2) are used, or as variable if terminals (1, and 3 or 2, and 3) are used. It can also be used as voltage divider when all three terminals are used provided that extreme terminals; i.e. 1, and 2 are connected to VCC, and GND. The output is taken at terminal 3.



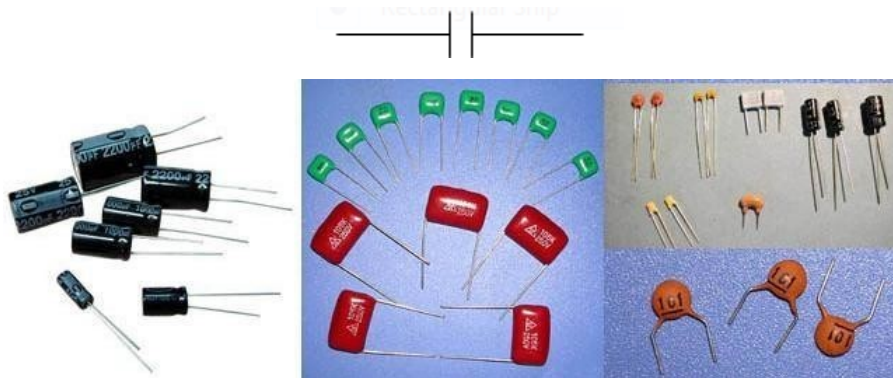
3. CAPACITOR:

A capacitor (originally known as a condenser) is a passive two-terminal electrical component used to store energy electrostatically in an electric field. By contrast, batteries store energy via chemical reactions. The forms of practical capacitors vary widely, but all contain at least two electrical conductors separated by a dielectric (insulator); for example, one common construction consists of metal foils separated by a thin layer of insulating film. Capacitors are

widely used as parts of electrical circuits in many common electrical devices.

Capacitors are also very commonly used. A lot have their values printed on them, some are marked with 3-digit codes, and a few are color coded. The same resources listed above for resistors can also help you identify capacitor values. They are typically marked with an “C” on a circuit board.

Symbol:



Applications:

- A. In tuned circuits.
- B. As bypass capacitors to by pass ac through it.
- C. Blocking capacitor to block dc components.

4. INDUCTORS:

An inductor, also called a coil or reactor, is a passive two-terminal electrical component which resists changes in electric current passing through it. It consists of a conductor such as a wire, usually wound into a coil. When a current flows through it, energy is stored in a magnetic field in the coil. When the current flowing through an inductor changes, the time-varying magnetic field induces a voltage in the conductor, according to Faraday’s law of electromagnetic induction, which by Lenz’s law opposes the change in current that created it.

Inductors, also called coils, can be a bit harder to figure out their values. If they are color coded, the resources listed for resistors can help, otherwise a good meter that can measure inductance will be needed. They are typically marked with an “L” on a circuit board.



Applications:

- A. Filter chokes for smoothing and pulsating currents produced by rectifiers.
- B. Audio frequency chokes, to provide high impedance at audio frequencies.

5. GANGED CONDENSERS:

In tuning circuits it is desired to change the value of capacitance readily, this is done by means of variable capacitors. The most common variable capacitor is the air ganged capacitor. By rotating the shaft, we can change the distance between movable and fixed sets of plates like that capacitance value is changed.

Applications:

Used in tuning circuits

6. RELAYS:

Switches closes its contacts by the mechanical activation of its lever, the relay do this by an electromagnetic coil pulling its contacts current and wound over a core of soft magnet. The moving armature of core causes the contact closure.



Relays can have several poles and several contacts. Types of Relays are:

- A. Reed Relays
- B. Solid State Relays
- C. Over load relays etc.

Applications:

In telephone networks

7. BREAD BOARDS:

This is the platform (or chasis) on which any circuit can be ringed up to provide inter connections between electronics components and devices.

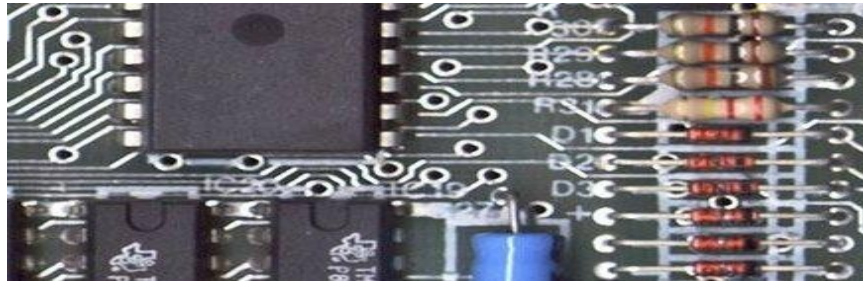
The advantage of bread board is, the components can be connected (or) disconnected easily. It has holes both horizontally and vertically as shown in the figure.

The horizontal holes at the top and bottom are having internal shorts where as in the remaining part vertical holes are shorted internally.



8. PCB (PRINTED CIRCUIT BOARD):

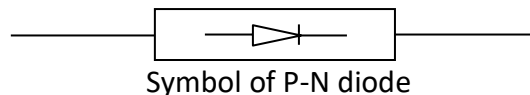
A printed **circuit board (PCB)** mechanically supports and electrically connects electronic components using conductive tracks, pads and other features etched from copper sheets laminated onto a non-conductive substrate. Components (e.g. capacitors, resistors or active devices) are generally soldered on the **PCB**.



9. DIODE:

A popular semiconductor device called a diode is made by combining P & N type semiconductor materials. The doped regions meet to form a P-N junction. Diodes are unidirectional devices that allow current to flow through them in one direction only.

The schematic symbol for a semiconductor diode is shown in fig-1. The P-side of the diode is called the anode (A), while the N-side of the diode is called the cathode (K).



Symbol of P-N diode

Specifications:

A. Breakdown voltage rating V_{BR}

The Breakdown voltage rating V_{BR} is the voltage at which avalanche occurs. This rating can be designed by any of the following: Peak Inverse Voltage (PIV); Peak Reverse Voltage (PRV); Break down voltage rating V_{BR}

B. Average forward – current rating, I_F

This important rating indicates the maximum allowable average current that the diode can handle safely, the average forward current rating is usually designated as I_F . Maximum reverse current, IR1N4007 silicon diode specifies a typical I_R of 0.05 A for a diode junction.

Temperature T_J of 250C and a reverse voltage V_R of 100 V. The maximum rating of a diode should never be exceeded.

Testing of diode:

Using an ohmmeter to check a diode: when using an ohm meter, check the resistance of the diode in one direction, then reverse the meter leads and measure the resistance of the diode in the other direction. If the diode is good it should measure a very high resistance in one direction, and a low resistance in the other direction. For a silicon diode the ratio of reverse resistance R_R , to forward resistance R_F should be very large, such as 1000:1 or more.

Note: If the diode is shorted it will measure a low resistance in both the directions. If the diode is open, it will measure a high resistance in both the directions.

Using a DMM to check a diode: Most digital multimeters provide a special range for testing the diodes. This range is called the diode range. This is the only range setting on the DMM that can provide the proper amount of forward bias for the diode being tested. It is important to note that when the digital multimeter forward biases the diode being tested, the digital

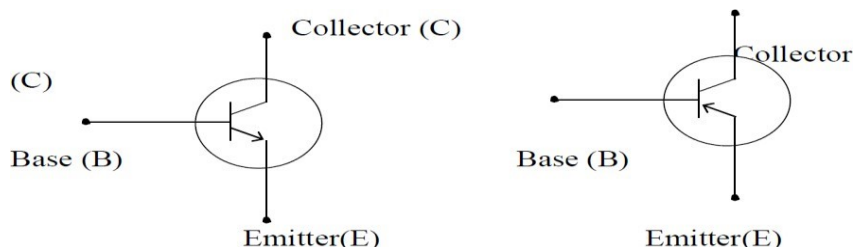
A good silicon diode tested with the Identification:



- A. Rectifiers, Clippers and Clampers.
B. Signal detector.
C. Digital logic gates.

In npn transistors, the majority current carriers are free electrons in the emitter and collector, while the majority current carriers are holes in the base. The opposite is true in the pnp transistor where the majority current carriers are holes in the emitter and collector, and the majority current carriers are free electrons in the base.

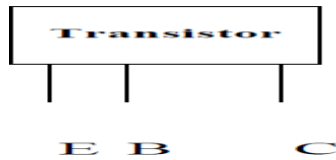
1. Emitter 2. Base 3. Collector



Schematic symbols for transistors (a) npn transistor (b) pnp transistor.

In order for a transistor to function properly as an amplifier, the emitter-base junction must be forward biased and the collector base junctions must be reverse biased.

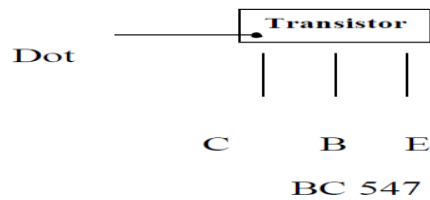
There are three leads in a Transistor called collector, emitter and base. When a transistor is to be connected in a circuit it is necessary to identify the leads of transistor before connecting in a circuit. The identification of the leads of transistor varies with manufacturer. There are three systems in general.



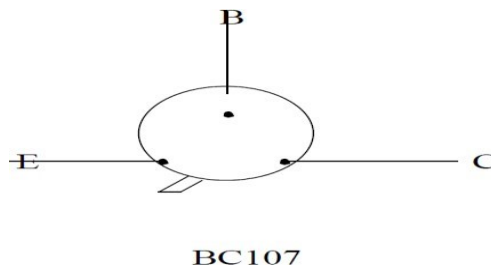
When the lead of a transistor is in the same plane and unevenly as in above fig., they are identified by the position and spacing of leads. The central lead is the base lead. The collector lead is identified by the large spacing existing between it and the base lead. The remaining is the emitter.

When the leads of a transistor are in the same plane but evenly spaced, the central lead is the base, the lead identified by dot is the collector and the remaining lead is the emitter.

When the leads of a transistor are spaced around the circumference of a circle, the three leads



are generally in E-B-C order clockwise from a notch.



Specifications:

In all cases, the maximum ratings are given for collector-base voltage, collector emitter voltage, emitter base voltage, collector current and power dissipation.

- **Power dissipation rating P_d (Max):**

The product of V_{CE} and I_C gives the power dissipation, P_d of the transistor. The product of $V_{CE} \times I_C$ must not exceed the maximum power dissipation rating, P_d (Max) of the transistor is nearly 1Watt.

- **Derating factor:**

Manufacturers usually supply derating factors for determining the power dissipation rating at any temperature above 25°C . The derating factor is specified in $\text{Watt}/^\circ\text{C}$. For

example if a transistor has a derating factor of 2 mW/°C, then for each 10 °C rise in junction temperature the power rating of the transistor is reduced by 2 mW.

- **Breakdown voltage ratings:**
- A data sheet lists the breakdown voltage ratings for the emitter- base, collector-base, and collector-emitter junctions. Exceeding these voltage ratings can destroy the transistor.
BVCEO is 60V, BVCEO is 40V and BVEBO is 6V.

Testing of BJTs:

- **Checking a transistor with an ohmmeter:**

To check the base-emitter junction of an npn transistor, first connect the ohmmeter and then reverse the ohmmeter leads. The resistance indicated by the ohmmeter should be low since the base emitter junction is forward biased. The resistance indicated by the ohmmeter should read high because the base emitter junction is reverse biased. For a good pn junction made of silicon the ratio R_R/R_F should be equal to or greater than 1000:1.

To check the collector-base junction, repeat the process described for the base-emitter junction.

- **Shorted and open junctions:**

A low resistance across the junction in both directions implies that the emitter-base or collector-base junctions are shorted. If the ohmmeter indicates a high resistance in both directions, then the junctions are open. In both cases the transistor is defective and must be replaced.

- **Checking a transistor with a Digital Multimeter (DMM):**

- Insert the transistor in the provided slots, position the knob of DMM in hFE mode and check the hFE value.

Applications:

- A. Amplifiers.
- B. Oscillators.
- C. Switches.

11. Field Effect Transistors (FETs):

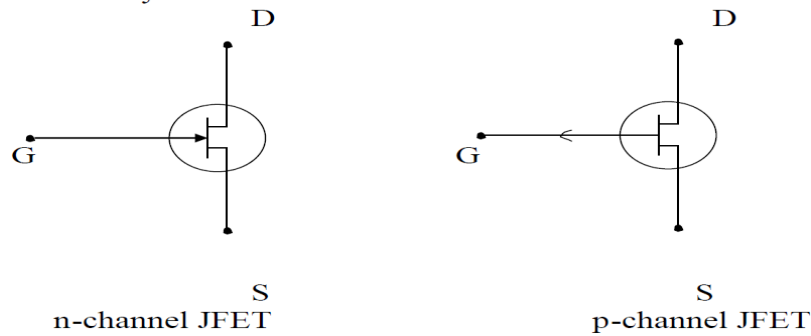
The field effect transistor (FET) is a three terminal device similar to the bipolar junction transistor. The FET, however, is a unipolar device, which depends on only one type of charge carriers; either electrons or holes. There are basically two types of FETs. The junction field effect transistor, abbreviated JFET, and the metal oxide semiconductor field effect transistor, abbreviated MOSFET.

A junction field effect Transistor is a three terminal semiconductor device in which current conduction is by one type of carriers i.e., electrons or holes.

There are two basic types of FET's

1. Junction field effect transistor (JFET)
2. Metal oxide field effect transistor (MOSFET)

Schematic Symbol of JFET:



Specifications:

- A. A.C. drain resistance (r_d): Drain resistance has a large value, ranging from $10\text{k}\Omega$ to $1\text{M}\Omega$
- B. Transconductance (g_m): It is expressed in mA/V
- C. Amplification factor (μ): It is product of A.C drain resistance and transconductance.

S.No	Parameters	BFW10	BFW11
1	I_{DSS}	8-20mA	4-10mA
2	V_{GS}	4-8V	2-4V
3	g_m	3.2mA/V	3.2 mA/V

Pinch off voltage:

The value V_p is the start of the interval V_p to $V_{DS\text{ max}}$ during which I_D remain constant. As V_{DS} is increased from 0V to V_p called the pinch off voltage, I_D increases from 0 to the maximum drain current that can be attained without destroying the JFET, the voltage I_{DSS} .

V_{GS} (off): Maximum drain current flows when $V_{GS} = 0$ and minimum drain current when $V_{GS} = V_{GS}(\text{off})$.

Testing:

In case of FET, drain to source should be a fixed resistance in either direction. Gate to drain or gate to source should be an open circuit or a very high resistance.

Applications:

Used in tuners of radio and TV receivers Amplifiers and Voltage variable resistor.

12. MOSFETs:

The metal-oxide semiconductor field effect transistor has a gate, source and drain just like JFET. Like a JFET, the drain current in a MOSFET is controlled by the gate-source voltage V_{GS} . There are two basic types of MOSFETs. The enhancement type and depletion type. The enhancement type MOSFET is usually referred to as an E-MOSFET and the depletion type MOSFET is referred to as a D-MOSFET.

The key difference between JFETs and MOSFETs is that the gate terminal in a MOSFET is insulated from the channel. Because of this, MOSFETs are sometimes referred to as insulated gate FETs or IGFETs. Because of the insulated gate, the input impedance of a MOSFET is many times higher than that of a JFET.

Types of MOSFET'S:

- A. n-channel D- MOSFET
- B. p-channel D-MOSFET
- C. n-channel E-MOSFET
- D. p-channel E-MOSFET

Specifications:

A typical MOSFET is the 3N200 made by BEL. It has two independent gates against only one in a common MOSFET. Its specifications are drain to source voltage $V_{DS} = 0.2V$ to $20V$

Gate 1 to source voltage $V_{G1S} = 0.6 V$ to $+3 V$

Gate 2 to source voltage $V_{G2S} = 0.6 V$ to $+6 V$ Drain to gate 2 voltage $V_{DG2} = +20V$ Drain current $I_D = 50 mA$

Transistor dissipation $PT = 330 mw$ Derating = $2.2 m2/0C$

Testing:

1. In case of MOSFET, drain to source should be a fixed resistance in either direction
2. Gate to drain or gate to source should be an open circuit or a very high resistance (greater than FET).

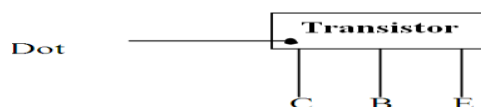
The device under test in the given circuit is a depletion type N- channel JFET, with the gate circuit kept open, the magnitude of the drain current is sufficient to make the $I_D R_2$ drop large enough. So that the BJT is forward biased and driven into its ON state. Therefore the lamp glows. The switch SW is now closed. The bias on the FET gate then causes depletion of its channel.

This lowers the $I_D R_2$ drop to the point where conduction through the BJT output circuit fails to keep the bulb glowing. All this will happen if the FET is in good condition. On the other hand, a short circuited FET will keep the lamp On in either position of switch SW, while an open FET will fail to switch the indicator lamp ON.

Power Transistors:

The two types of power transistors made with of the alloy junction type and the silicon planar type. However, large power means a high current circulating through the device, requiring bigger areas for a given current density, the active emitter and collector areas can be over 10 .

Identification:



SL100

Specifications:

One way to distinguish transistors is by the number marked on them. Thus low frequency, low power transistors bears the numbers AC125, AC126, BC147 and BC148. An example of a high frequency, low power transistor is BF115; power transistors are BD138, AD149, 2N3055 etc. Similarly, germanium and silicon transistors are distinguished by the first letters A and B respectively.

Type	BVcbo	BVceo	BVebo	Ic	Maximum Collector Dissipation at 25° C	β	Vce	Ic	Icbo
2N696 Medium level	60	40	5	0.3	0.6	20-60	10	150	1

BVCEO: maximum value of voltage across collector-emitter circuit with base open.

BVCBO: maximum value of voltage across collector-base circuit with emitter open.

BVEBO: maximum value of voltage across emitter-base circuit with collector open.

ICmax: maximum value of DC collector current.

PT: max. power dissipation in absence of heat sinks at ambient temperature.

FT: gain bandwidth ICBO: leakage current.

Testing:

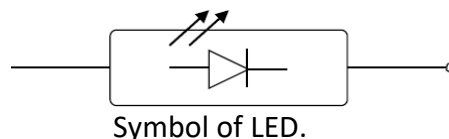
Follow the same procedure as the ordinary BJT-testing using multimeter.

The characteristics of a BJT under different configurations can be observed directly on a CRT screen. The transistors leads are simply inserted in the three sockets of a device called a curve tracer.

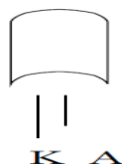
13. Light Emitting Diodes LEDs:

As opposed to other diodes that give off heat when conducting, LEDs emit light. In the latter, the recombination of charge carriers across the PN junction releases optical energy when the electrons fall from the conduction to the valence band. The heat emission is negligible in light emitting materials like gallium arsenic phosphide and gallium phosphide.

LEDs must, of course, be covered in a transparent or translucent material. The wavelength of the radiation for a given color is given by the relation $\lambda = 1.23/E_g$. Where E_g is the energy gap between conduction and valence bands. Its value is 1.45 eV for GaAs, 3 eV for GaAsP and 2.25 eV for GaP. The colors obtained from these materials are red, yellow and green respectively.



Identification:



Longer terminal is cathode and the other is Anode.

Specifications:

- VF operating : 1.3 v to 2.5 v
- Forward voltage (max): 5 v.
- Forward current 5 mA to 15 mA
- Reverse breakdown : 10 v to 12 V
- Operating life: 100,000 hours.
- Turn on time: 10-20 nsec.
- Turn off time: 80-100 nsec.

Testing:

The diode is simply put across a multimeter to see if the reading is different when the polarity is reversed. The LED will glow and show a resistance between 30 ohms and 50 ohms when forward biased.

❖ **Lab Reports**

The following format should be adhered to by the students in all their laboratory reports:

☒ **Objective**

This should state clearly the objective of the experiment. It may be the verification of law, a theory or the observation of particular phenomena. Writing out the objective of the experiment is important to the student as it emphasizes the purpose for which the experiment is conducted.

☒ **Brief Theory**

The related theory of the experiment must be discussed **briefly**. It assists the student in making a conclusion based on comparison between the experimental results to the theory.

☒ **Results**

All experimental results which have been approved by the lab instructor (including graphs) must be attached in the report.

☒ **Discussion & Conclusion**

The student must form some elaboration on the results of his analysis. Usually this involves deducing whether the final results show that the aim of the experiment has been achieved or not.

EXPERIMENT ONE

PN JUNCTION DIODE CHARACTERISTICS

1.1 Objectives:

To plot the V-I characteristics of a PN junction diode in both forward and reverse directions. Find cut in voltage (knee voltage), static and dynamic resistance in forward direction at forward current of 2mA & 8mA respectively. Find static and dynamic resistance at 10V in reverse bias condition.

1.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1.	Semiconductor diode trainer Board Containing: DC Power Supply Diode (Silicon) Diode (Germanium) Carbon Film Resistor	(0-15) V 1N 4007 OA79 1 K Ω , 1/2 W	1 1 1 1
2.	DC Voltmeter	(0-20) V	1
3.	DC Ammeter	(0-20) mA	1
4.	Connecting wires	5A	10

1.3 Theory:

A p-n junction diode conducts only in one direction. The V-I characteristics of the diode are curve between voltage across the diode and current through the diode. When external voltage is zero, circuit is open and the potential barrier does not allow the current to flow. Therefore, the circuit current is zero. When P-type (Anode is connected to +ve terminal and n- type (cathode) is connected to -ve terminal of the supply voltage, is known as forward bias. The potential barrier is reduced when diode is in the forward biased condition. At some forward voltage, the potential barrier altogether eliminated and current starts flowing through the diode and also in the circuit. The diode is said to be in ON state. The current increases with increasing forward voltage. When N-type (cathode) is connected to +ve terminal and P-type (Anode) is connected -ve terminal of the supply voltage is known as reverse bias and the potential barrier across the junction increases. Therefore, the junction resistance becomes very high and a very small current (reverse saturation current) flows in the circuit. The diode is said to be in OFF state. The reverse bias current due to minority charge carriers.

1.4 Procedure:

A. Forward Bias:

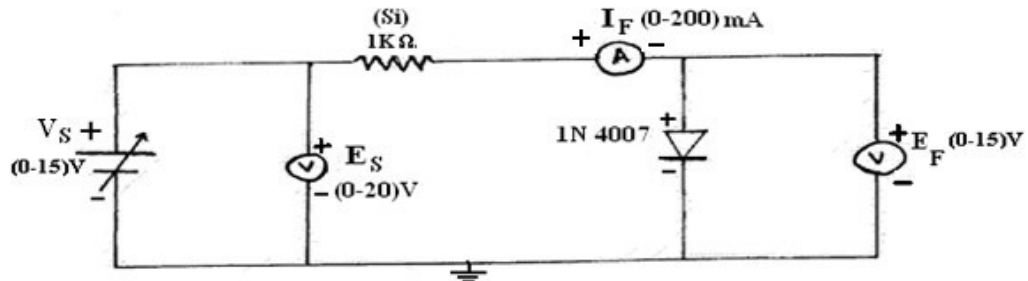
1. Connect the circuit as shown in figure (1).
2. Vary the supply voltage E_s in steps and note down the corresponding values of E_f and I_f as shown in the tabular column.

B. Reverse Bias:

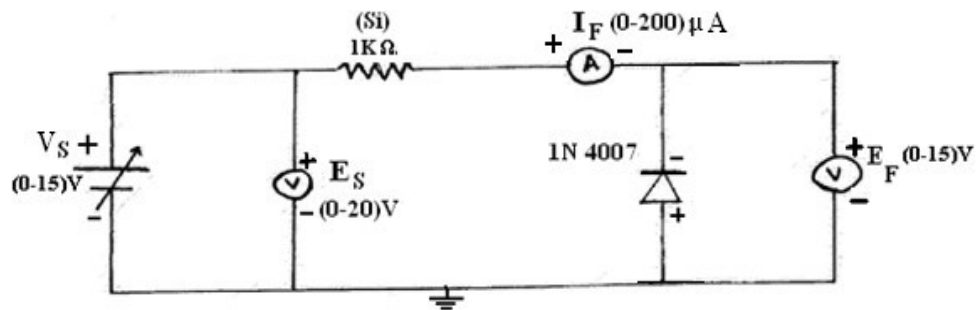
1. Connect the circuit as shown in figure (2).
2. Repeat the procedure as in forward bias and note down the corresponding Values of E_r and I_r as shown in the tabular column.

1.5 Circuit Diagrams:

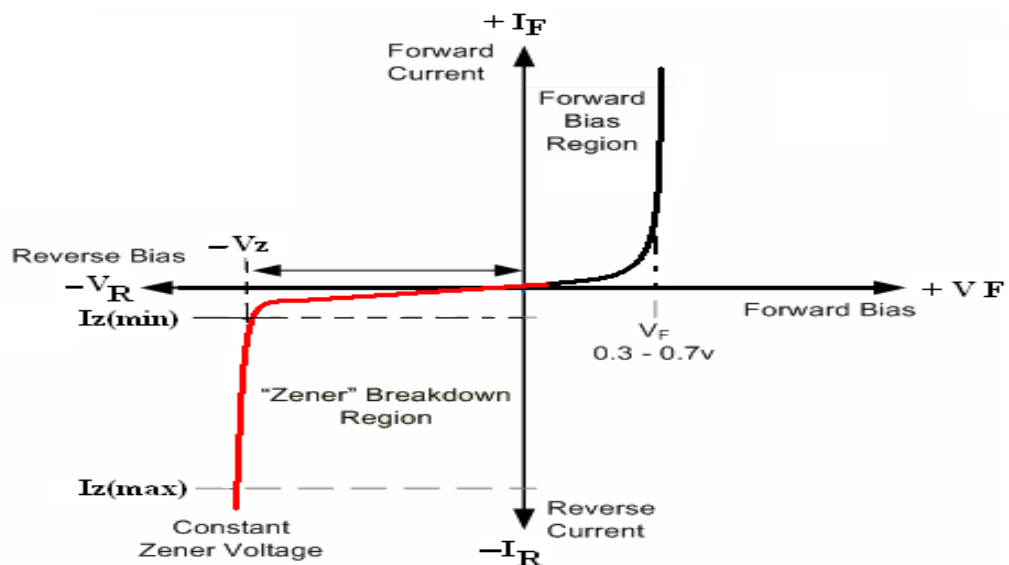
Forward Bias



Reverse Bias



1.6 Expected Graphs:



1.7 Tabular Column:

Forward Bias			Reverse Bias		
E_s (volts)	E_f (volts)	I_f (mA)	E_s (volts)	E_r (volts)	I_r (μ A)
0.1			0.1		
0.2			0.2		
0.3			0.3		
0.4			0.4		
0.5			0.5		
0.6			0.6		
0.7			0.7		
0.8			0.8		
0.9			0.9		
1			1		
2			2		
4			4		
6			6		
8			8		
10			10		
12			12		
14			14		

1.8 Precautions:

1. Ensure that the polarities of the power supply and the meters as per the circuit diagram.
2. Keep the input voltage knob of the regulated power supply in minimum position both when switching ON or switching OFF the power supply.
3. No loose contacts at the junctions.
4. Ensure that the ratings of the meters are as per the circuit design for precision.

1.9 Calculations:

A. Forward Bias:

Static Resistance at 8mA = $E_f / I_f =$

Static resistance at 2mA = $E_f / I_f =$

Dynamic resistance at 8mA = $\Delta E_f / \Delta I_f =$

Dynamic resistance at 8mA = $\Delta E_f / \Delta I_f =$

B. Reverse Bias:

Static Resistance at (10V) = $E_r / I_r =$

Dynamic resistance at (10V) = $\Delta E_r / \Delta I_r =$

1.10 Questions:

1. Define depletion region of a diode?
2. What is meant by transition & space charge capacitance of a diode?
3. Is the V-I relationship of a diode Linear or Exponential?
4. Define cut-in voltage of a diode and specify the values for Si and Ge diodes?
5. What are the applications of a p-n diode?
6. Draw the ideal characteristics of P-N junction diode?
7. What is the diode equation?
8. What is PIV?
9. What is the break down voltage?
10. What is the effect of temperature on PN junction diodes?
11. Specifications of diodes

1.11 Lab Assignment:

To plot the V-I characteristics of a PN junction (Germanium) diode in both forward and reverse directions by using multisim.

1.12 Result:

V-I characteristics of PN junction are plotted and verified in both forward and reverse directions.

Forward direction

- i. Cut-in-voltage=0.7V
- ii. a) Dynamic Resistance (at 8 mA) =
b) Dynamic Resistance (at 2mA) =
- iii. a) Static Resistance (at 8mA) =
- iv. b) Static Resistance (at 2mA) =

Reverse Direction

- i. Static Resistance (at 10V) =
- ii. Dynamic Resistance (at 10 V) =

EXPERIMENT TWO

APPLICATIONS OF DIODES

2.1 Objectives:

To verify the truth table for Logic Gates (AND & OR) using Diodes.

2.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
01	Diode	IN4007	02
02	Resistance	1K Ω ,10K Ω	01
03	Regulated power supply	(0-30V)	01
04	Voltmeter/Digital Multimeter		01
05	Breadboard and Wires		

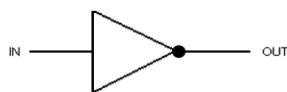
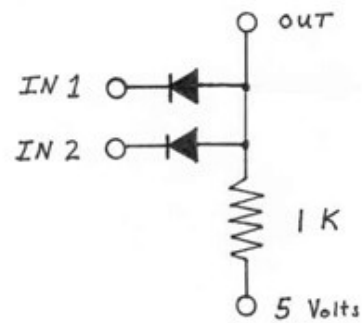
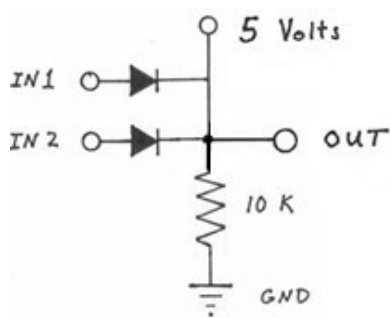
2.3 Theory:

There are three types of basic Logic Operations. These can be executed by electronic circuits or devices (also called gates or switches). They are: AND, OR and NOT. AND and OR have two inputs and one output. NOT has one input and one output. Using 0's to represent 0 Volts and 1's to represent 5 Volts, the tables below show the outputs for all the possible inputs on the logic gates. Below each table (called a "truth table") is the electronic symbol for the device

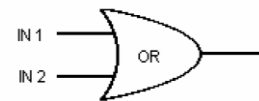
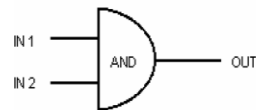
2.4 Procedure:

1. Connections are made as per the circuit diagram of AND gate.
2. Apply the supply of 5V
3. Give the In1 and In2 to the circuit according to the truth table (i.e for „0“ apply 0V and for „1“ apply 5V) and verify the output of the circuit and tabulate the readings.
4. Verify the truth table with the truth table of the AND gate.
5. Repeat the steps 1 to 4 for the circuit diagram of OR gate.

2.5 Circuit Diagrams:



DDL OR Gate



DDL AND Gate

Truth Table:

In1	In2	Output
0	0	0
0	1	1
1	0	1
1	1	1

In1	In2	Output
0	0	0
0	1	0
1	0	0
1	1	1

2.6 Tabular Column:

Logic Gate	In1	In2	O/P
AND			

Logic Gate	In1	In2	O/P
OR			

2.7 Precautions

1. Loose and wrong connections should be avoided.
2. Supply should be switched on only after giving all the input connections.
3. Power should be switched off while connecting.

2.8 Questions:

1. Explain about the switching action of transistor
2. Draw the internal diagram of PN diode
3. Give some applications of diode.

2.9 Lab Assignment:

To verify the Truth Table.

2.10 Result:

The truth tables of the AND and OR gates using DDL logic is verified.

EXPERIMENT THREE

HALFWAVE RECTIFIERS WITH/WITHOUT FILTERS

3.1 Objectives:

Examine the input and output waveforms of a half wave rectifier without and with filters. Calculate the ripple factor with load resistance of 500Ω , $1\text{ K}\Omega$ and $10\text{ K}\Omega$ respectively. Calculate ripple factor with a filter capacitor of $100\mu\text{F}$ and the load of $1\text{K}\Omega$, $2\text{K}\Omega$ and $10\text{K}\Omega$ respectively.

3.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1	Rectifier and Filter trainer Board Containing a) AC Supply. b) Silicon Diodes c) Capacitor	(9-0-9) V	1
		1N 4007	1
		$0.47\mu\text{F}$	1
2	a) DC Voltmeter	(0-20) V	1
	b) AC Voltmeter	(0-20) V	1
3	DC Ammeter	(0-50) mA	1
4	Cathode Ray Oscilloscope	(0-20) MHz	1
5	Decade Resistance Box	10Ω - $100\text{K}\Omega$	1
6	Connecting wires	5A	12

3.3 Theory:

During positive half-cycle of the input voltage, the diode D1 is in forward bias and conducts through the load resistor R1. Hence the current produces an output voltage across the load resistor R1, which has the same shape as the +ve half cycle of the input voltage.

During the negative half-cycle of the input voltage, the diode is reverse biased and there is no current through the circuit. i.e, the voltage across R1 is zero. The net result is that only the +ve half cycle of the input voltage appears across the load. The average value of the half wave rectified o/p voltage is the value measured on dc voltmeter.

For practical circuits, transformer coupling is usually provided for two reasons.

1. The voltage can be stepped-up or stepped-down, as needed.
2. The ac source is electrically isolated from the rectifier. Thus preventing shock hazards in the secondary circuit.

3.4 Procedure:

Half Wave Rectifier without filter

1. Connect the circuit as shown in figure (a).
2. Adjust the load resistance, R_L to 500Ω , and note down the readings of input and output voltages through oscilloscope.
3. Note the readings of dc current, dc voltage and ac voltage.
4. Now, change the resistance the load resistance, R_L to $1\text{ K}\Omega$ and repeat the procedure as above. Also repeat for $10\text{ K}\Omega$.
5. Readings are tabulated as per the tabular column.

Half Wave Rectifier with filter

1. Connect the circuit as shown in figure (b) and repeat the procedure as for half wave rectifier without filter.

3.5 Circuit Diagrams:

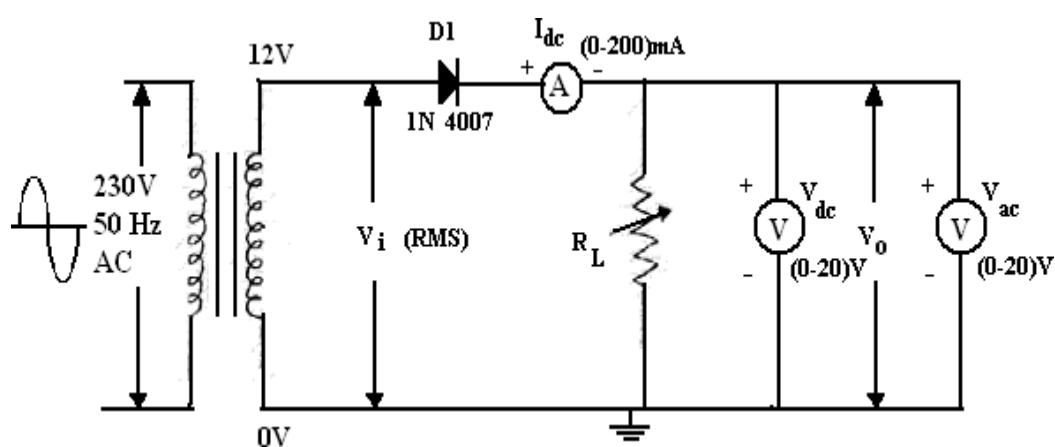


Figure (a): Half Wave Rectifier without Filter

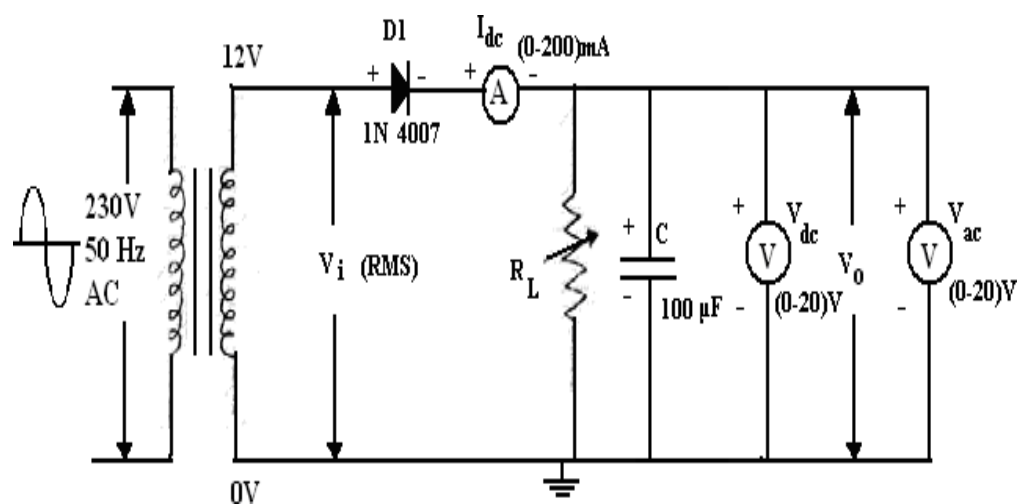
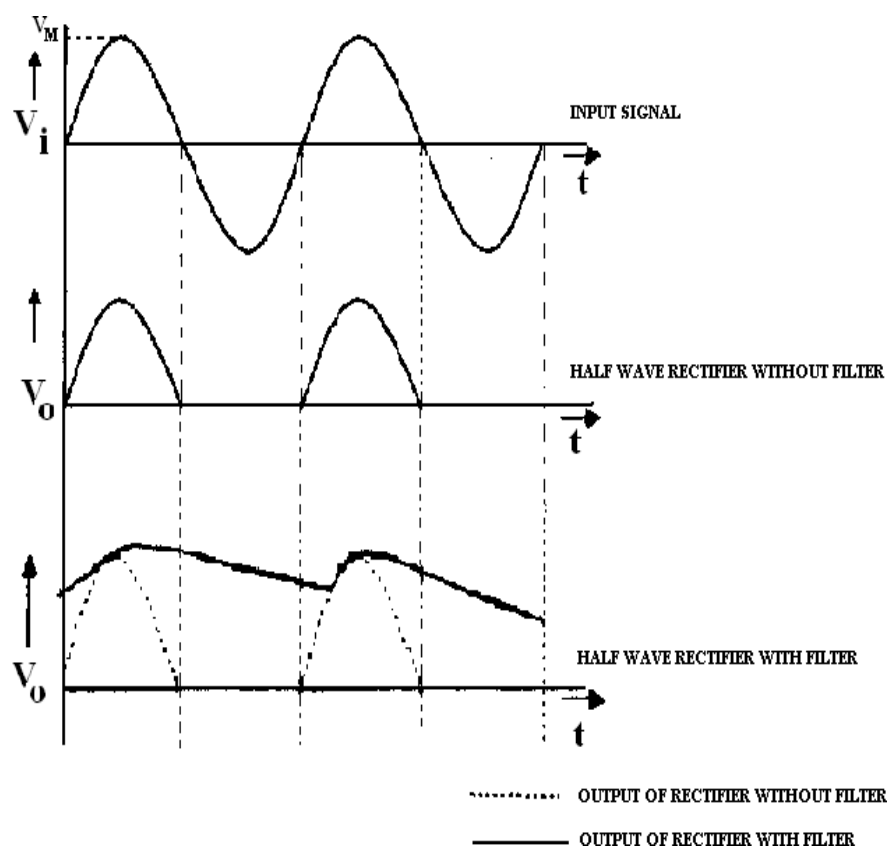


Figure (b): Half Wave Rectifier with Filter

3.6 Expected Graphs:



3.7 Tabular Column:

Half Wave Rectifier without Filter

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Average Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac} / V_{dc}$
1.	500 Ω						
2.	1K Ω						
3.	10K Ω						

Half Wave Rectifier with Filter ($C=10\mu F$)

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Average Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	1K Ω						
2.	2K Ω						
3.	10K Ω						

3.8 Precautions:

1. No loose contacts at the junctions.
2. Meters of correct ranges must be used for precision

3.9 Questions:

1. What is a rectifier?
2. How Diode acts as a rectifier?
3. What is the significance of PIV? What is the condition imposed on PIV?
4. What is meant by regulation?
5. What is meant by time constant?
6. Draw the o/p wave form without filter?
7. Draw the o/p wave form with filter?
8. What is meant by ripple factor? For a good filter whether ripple factor should be high or low?
9. What happens to the o/p wave form if we increase the capacitor value?
10. What happens if we increase the capacitor value?

3.10 Lab Assignment:

Plot the wave forms of Half wave rectifier with $R_L=5000$ ohms, $C = 680\mu F$.

3.11 Result:

1. Input and Output waveforms of a half-wave with /without filter are observed and plotted.
2. For Half-wave rectifier without filter- γ , Ripple factor at 500 Ω =
1K Ω =
10 K Ω =
3. For Half-wave rectifier with filter:- γ , Ripple factor at 1K Ω , 100 μF =
2K Ω , 100 μF =
10 K Ω , 100 μF =

EXPERIMENT FOUR

FULLWAVE RECTIFIERS WITH/WITHOUT FILTER

4.1 Objectives:

- Examine the input and output waveforms of a full wave (center tapped) rectifier without and with filters. Calculate the ripple factor with load resistance of 500 Ω , 1 K Ω and 10 K Ω respectively.
- Calculate ripple factor with a filter capacitor of 100 μ F and the load of 1K Ω , 2K Ω and 10K Ω respectively.

4.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1	Rectifier and Filter trainer Board		
	a) AC Supply.	(9-0-9) V	1
	b) Silicon Diodes	1N 4007	2
	c) Capacitor	0.47 μ F	1
2	a) DC Voltmeter	(0-20) V	1
	b) AC Voltmeter	(0-20) V	1
3	DC Ammeter	(0-50) mA	1
4	Cathode Ray Oscilloscope	(0-20) MHz	1
5	Decade Resistance Box	10 Ω -100K Ω	1
6	Electrolytic Capacitor	100 μ F	1
7	Connecting wires	5A	12

4.3 Theory:

The circuit of a center-tapped full wave rectifier uses two diodes D1&D2. During positive half cycle of secondary voltage (input voltage), the diode D1 is forward biased and D2is reverse biased.

The diode D1 conducts and current flows through load resistor R_L . During negative half cycle, diode D2 becomes forward biased and D1 reverse biased. Now, D2 conducts and current flows through the load resistor R_L in the same direction. There is a continuous current flow through the load resistor R_L , during both the half cycles and will get unidirectional current as show in the model graph. The difference between full wave and half wave rectification is that a full wave rectifier allows unidirectional (one way) current to the load during the entire 360 degrees of the input signal and half-wave rectifier allows this only during one half cycle (180 degree).

4.4 Procedure:

Full-wave Rectifier without filter

1. Connect the circuit as shown in the figure (a).
2. Adjust the load resistance R_L to 500Ω and connect a capacitor of $100\mu F$ value in parallel with the load and note the readings of input and output voltages through Oscilloscope.
3. Note the readings of DC current, DC voltage and AC voltage.
4. Now change the load resistance R_L to 2000Ω and repeat the procedure as the above.
5. Readings are tabulate as per the tabular column.

Full-wave Rectifier with filter

1. Connect the circuit as shown in the figure (b).
2. Adjust the load resistance R_L to $1K\Omega$ and connect a capacitor of $100\mu F$ values in parallel with the load and note the readings of input and output voltages through Oscilloscope.
3. Note the readings of DC current, DC voltage and AC voltage.
4. Now change the load resistance R_L to $2K\Omega$ and repeat the procedure as the above. Also repeat for $10K$, $100\mu F$ values.
5. Readings are tabulate as per the tabular column.

4.5 Circuit Diagrams:

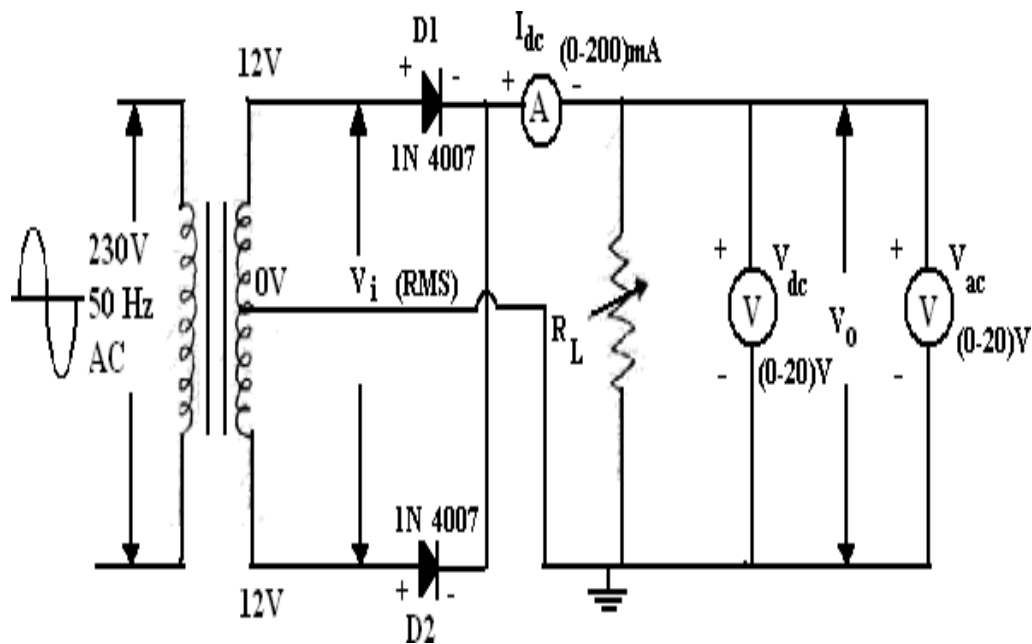


Figure (a): Full Wave Rectifier (Center-tap) Without Filter

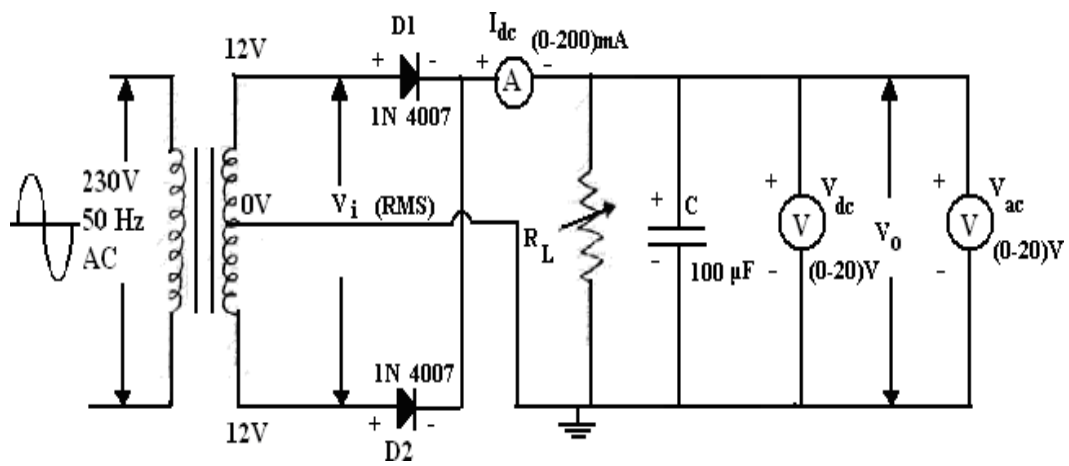
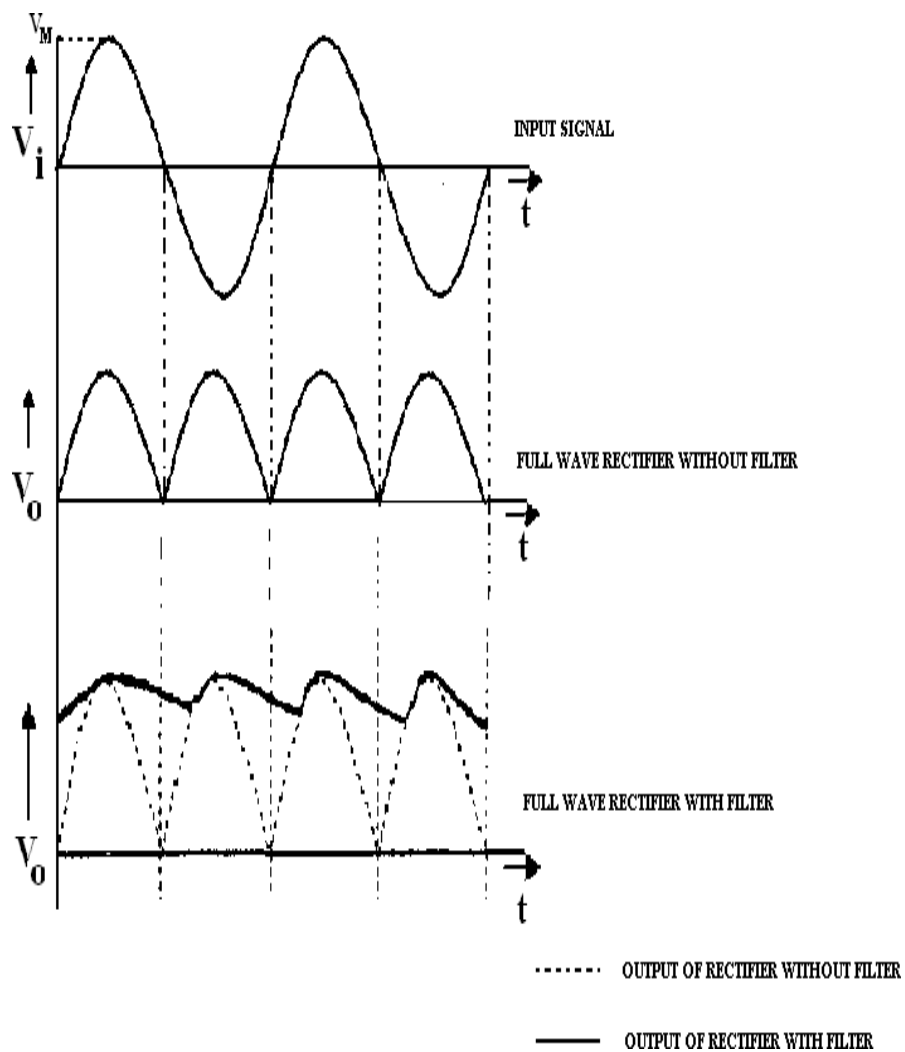


Figure (b): Full Wave Rectifier (center-tap) With Filter

4.6 Expected Graphs:



4.7 Tabular Column:

Full wave Rectifier (Center-tap) Without Filter

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Average Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	500 Ω						
2.	1K Ω						
3.	10K Ω						

Full wave Rectifier (Center-tap) With Filter $C = \text{---} \mu\text{F}$

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Average Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	500 Ω						
2.	1K Ω						
3.	10K Ω						

4.8 Precautions:

1. No loose contacts at the junctions.
2. Meters of correct range must be used for precision.

4.9 Questions:

1. What is a full wave rectifier?
2. How Diode acts as a rectifier?
3. What is the significance of PIV requirement of Diode in full-wave rectifier?
4. Compare capacitor filter with an inductor filter?
5. What is the theoretical maximum value of ripple factor for a full wave rectifier?
6. Draw the o/p wave form without filter? Draw the O/P? What is wave form with filter?
7. What is meant by ripple factor? For a good filter whether ripple factor should be high or low? What happens to the ripple factor if we insert the filter?
8. What is meant by regulation? Why regulation is poor in the case of inductor filter?
9. What is meant by time constant?
10. What happens to the o/p wave form if we increase the capacitor value? What happens if we increase the capacitor value?

4.10 Lab Assignment:

Plot the wave forms of full wave rectifier with $R_L=500\ \Omega$, $C = 470\ \mu\text{F}$.

4.11 Result:

1. Input and Output waveforms of a full-wave (center tapped) and bridge rectifier with /without filters are observed and plotted.
2. For Full-wave rectifier(center tapped) without filter- γ , Ripple factor at $500\ \Omega$, $100\ \mu\text{F} = 2\text{K}\Omega$, $100\ \mu\text{F} =$
3. For full-wave rectifier (Center tapped) with filter- γ , Ripple factor at $500\ \Omega$, $100\ \mu\text{F} = 2\text{K}\Omega$, $100\ \mu\text{F} =$

EXPERIMENT FIVE

THE DIODE LIMITER

5.1 Objectives:

To demonstrate the operation of various diodes Limiter circuits.

5.2 Components & Equipment required:

- Dual Trace Oscilloscope
 - DMM
 - Dual DC Power Supply.
 - Breadboard
 - Function Generator
- 2 IN 4001 Diodes
15K Ω Resistors
Jumpers.

5.3 Theory:

Clippers (Limiters) have the ability to limit "clip-off" a portion of the input signal without distorting the remaining part of the alternating waveform. The half-wave rectifier is an example of the series configuration of diode clipper. Depending on the direction of the diode, the positive or negative region of the input signal is "clipped" off. There are two general categories of clippers: *series* and *parallel*. The series configuration is defined as one where the diode is in series with the load, while the parallel variety has the diode in a branch parallel to the load. Figure (a) shows positive parallel limiter.

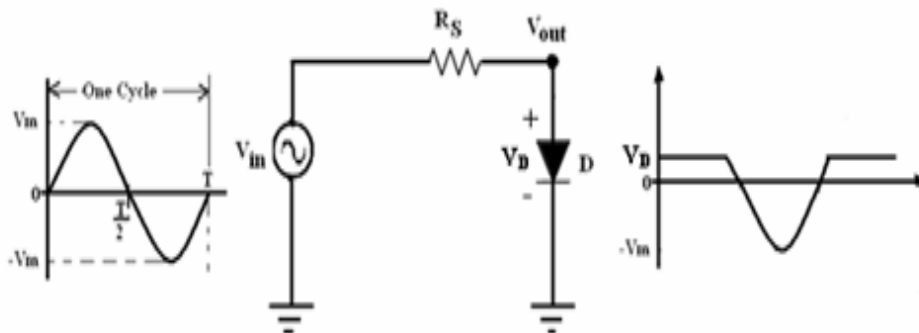


Figure (a): Parallel Limiter

The addition of a dc supply such as shown in Figure (b) can have a pronounced effect on the output of a clipper, and is known as biased limiter. V_{DC} in the voltage divider method is given by:

$$V_{DC} = \left(\frac{R_2}{R_1 + R_2} \right) * V_{CC}$$

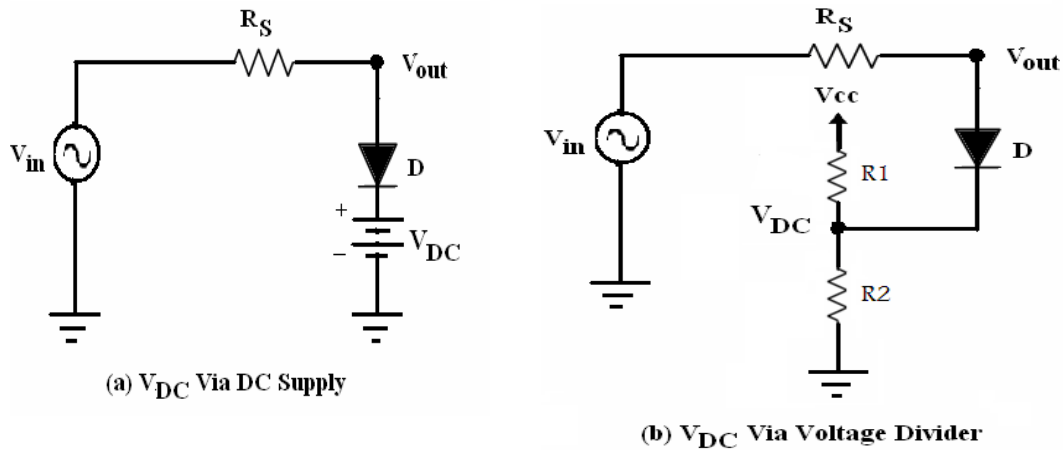


Figure (b): Biased Parallel limiters

5.4 PROCEDURE

1. Wire the circuit shown in Figure (c).
2. Connect CH1 at point A with respect to ground.
3. Adjust V_{in} to $6V_{p-p}/200\text{Hz}$ sine wave.
4. Connect CH2 at point B with respect to ground to measure the limiting level.
5. On a graph draw both input and output signals.
6. Wire the circuit of figure (d). Gradually increase V_{DC} and observe V_{out} . Explain your results and determine V_{DC} value that will retain unclipped V_{out} .

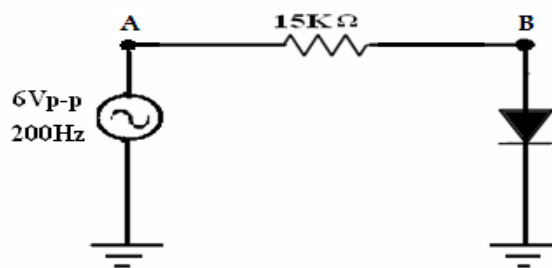


Figure (c): Basic Positive Limiter

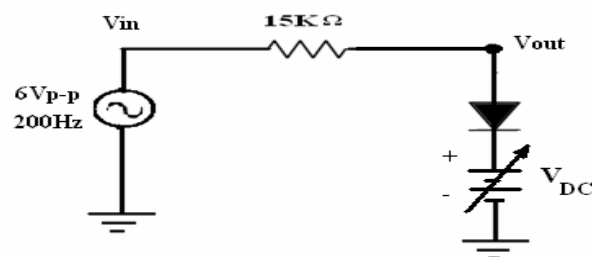


Figure (d): Biased Limiter

7. Wire the circuit of figure (e) and repeat steps 4 and 5.

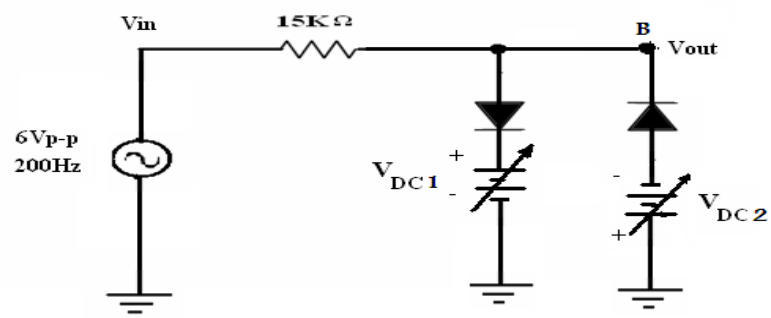


Figure (e): Double Biased Limiter

EXPERIMENT SIX

CLAMPERS AND VOLTAGE MULTIPLIER

6.1 Objectives:

To design Clampers and Voltage doubler for the given specification.

6.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1.	Diode (1N4007)	-	02
2.	Resistor & capacitors	As per design	-
3.	Multimeter	-	01

6.3 Procedure:

A. Clampers

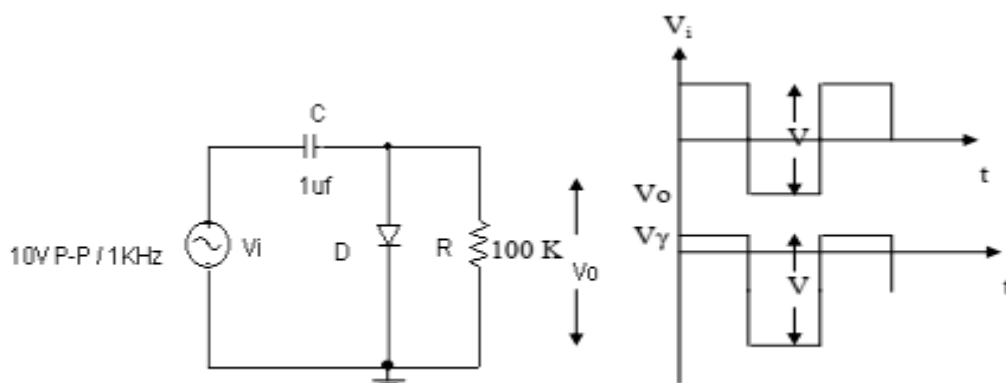
1. Connections are made as shown in the circuit diagram.
2. Apply a square wave / triangular wave / sine wave input of amplitude 10 V peak to peak and frequency of 1 kHz
3. Observe the input and output waveform.
4. Measure the clamping level and compare with the designed value

B. Voltage Multiplier

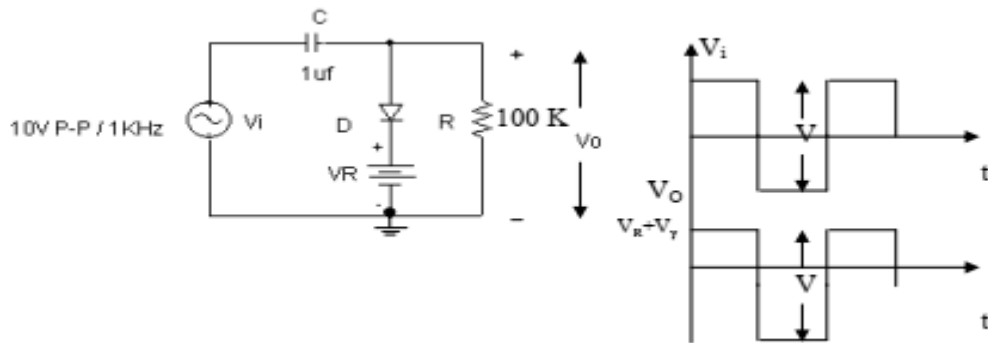
1. Connections are made as shown in the circuit diagram
2. Apply a square wave / triangular wave / sine wave input of amplitude 10 V peak to peak and frequency of 1 kHz
3. Observe the input and output waveform.
4. Measure the output voltage using multimeter in DC position.

Negative Clampers:

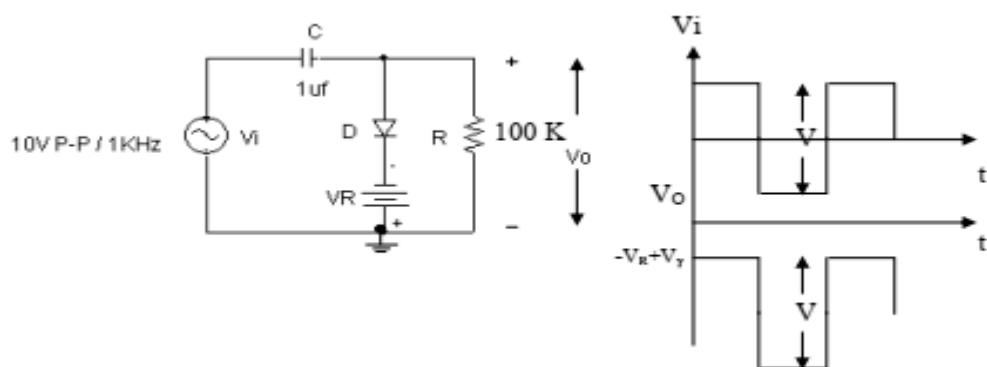
- Positive peak clamped to positive reference level ($V_R + V_T$)



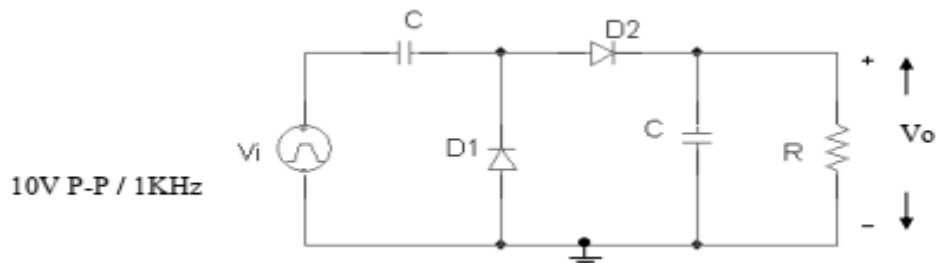
- Positive peak clamped to negative reference level ($-V_R + V_T$)



- Positive peak clamped to negative reference level ($-V_R + V_T$)



6.4 Circuit Diagram: Voltage Doubler



Design: Clampers

Example. 1: Design a clamping circuit to clamp the negative peak to +1V (Circuit 2)

$V_R - V_T = 1V$ Let the diode be silicon make then $V_T = 0.6V$

Then $V_R = 1 + 0.6 = 1.6V$

For a clamper $RC \gg T$, let $RC = 100 T$

Assume $f = 1 \text{ kHz}$, hence $T = 1 \text{ ms}$, choose $C = 1\mu f$

Then $R = 100 K\Omega$

Example. 2: Design a clamping circuit to clamp the positive peak to +2V (Circuit 3)

$V_R + V_T = 2V$ Let the diode be silicon make then $V_T = 0.6 V$

$V_R = 2 - 0.6$

$V_R = 1.4 V$

For a clamper $RC \gg T$, let $RC = 100 T$

Assume $f = 1\text{ kHz}$, hence $T = 1\text{ ms}$, choose $C = 1\mu\text{f}$ Then $R = 100\text{ k}\Omega$

Design: Voltage Doubler

For Voltage doubler, $RC \gg T$, let $RC = 100 T$

Assume $f = 1\text{ kHz}$, hence $T = 1\text{ ms}$, choose $C = 1\mu\text{f}$

Then $R = 100\text{ K}$

6.5 Result:

A. Positive Clampers			
#	Circuit	Designed Clamping Level	Observed Clamping Level
1.	Negative peak clamped to $-V_T$ level	$-V_T =$	
2.	Negative peak clamped to positive reference level ($V_R - V_T$)	$V_R - V_T =$	
3.	Negative peak clamped to negative reference level ($-V_R - V_T$)	$-V_R - V_T =$	
B. Negative Clampers			
1.	Positive peak clamped V_T level	$V_T =$	
2.	Positive peak clamped to positive reference level ($V_R + V_T$)	$V_R + V_T =$	
3.	Positive peak clamped to negative reference level ($-V_R + V_T$)	$-V_R + V_T =$	
C. Voltage Multiplier			
1.	Voltage Doubler	$V_o = 2V_M - 2V_T =$	

ZENER CHARACTERISTICS & ZENER AS A VOLTAGE REGULATOR

- To demonstrate the Zener diode characteristics in DC mode.
- To demonstrate the operation of Zener diode as a voltage regulator.

- **Function Generator**
- **Dual Trace Oscilloscope**
- **DMM (Digital Multi Meter)**
- **Breadboard**
- **Dual Dc power supply**

Zener Diodes are basically the same as the standard PN junction diode but are specially designed to have a low pre-determined reverse breakdown voltage as shown in figure (a) that takes advantage of this high reverse voltage. Zener diodes are used as voltage regulators.

It had been seen that the Zener diode has a region in its reverse bias characteristics of almost a constant negative voltage regardless of the value of the current flowing through the diode and remains nearly constant even with large changes in current as long as the Zener diodes current remains between the breakdown current $I_{Z(\min)}$ and the maximum current rating $I_{Z(\max)}$. The ability of controlling the voltage can be used to great effect to regulate or stabilize a voltage source against supply or load variations, and the function of a regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or the variation in the load current and the Zener diode will continue to regulate the voltage until the diodes current falls below the minimum $I_{Z(\min)}$ value in the reverse breakdown region.

Figure (a) is an analysis circuit for Zener diodes when are used as a voltage regulators. When E_S is less than V_Z , Zener acts as an open circuit and V_O is given by:

$$V_O = (R_L / (R_L + R_S)) E_S \quad (1)$$

And, $I_L = V_L / R_L$ (2)

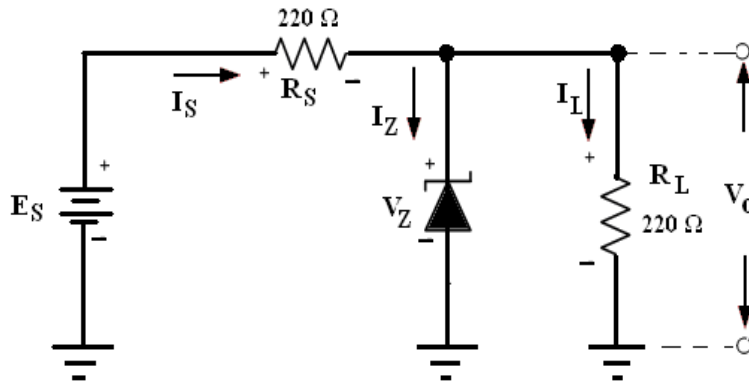


Figure (a): Analysis diagram for Zener diode as a regulator

When $E_S \geq V_Z$, then:

$$V_O = V_L = V_Z \quad (3)$$

$$\text{And, } I_S = I_Z + I_L \quad (4)$$

The source current (I_S) is given by:

$$I_S = (E_S - V_O) / R_S \quad (5)$$

And power dissipation (P_Z) is:

$$P_Z = I_Z V_Z \quad (6)$$

Regulation percentage (% V_O)

$$\%V_O = [(V_{NL} - V_{FL}) / V_{FL}] * 100\% \quad (7)$$

Where, V_{NL} — Output voltage without Load, and V_{FL} — Output voltage with Load

7.4 Procedure:

PART 1: ZENER CHARACTERISTICS

1. Wire the network of figure (b).

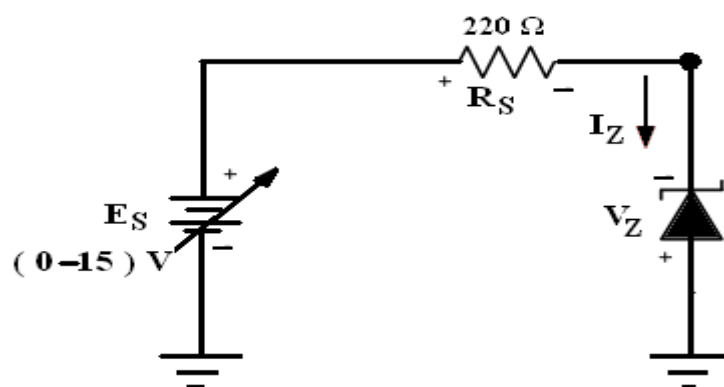


Figure (b): Zener Characteristics Circuit

2. Increase the dc supply voltage in small steps while simultaneously measuring the voltage across (V_Z) and calculate the current (I_Z) using ($I_Z = V_{RS} / R_S$) through the Zener diode. In the vicinity of the Zener's knee voltage (approximately 6 V), make these steps approximately 0.05V **Do not exceed a Zener current of 40 mA**. Record your data in Table 7.1.

3. Plot your results for the corresponding Zener current and voltage values on the graph provided for this purpose.
4. At $I_Z = 20\text{mA}$, determine R_Z ,
5. Wire the circuit shown in figure (c). Calculate I_Z , I_L and V_O . Record your results in table 4.2.

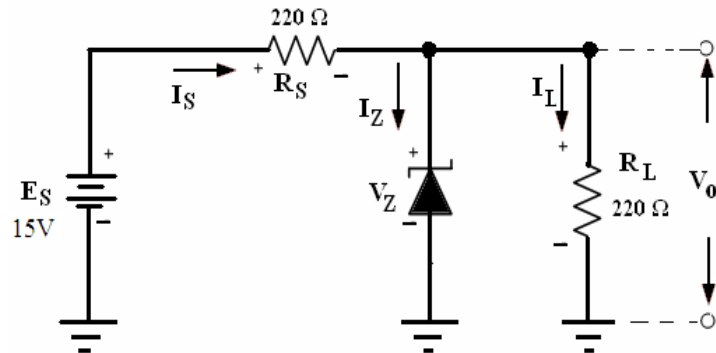


Figure (c): Analysis diagram for Zener diode as a regulator

6. Remove R_L , and calculate V_O .
7. Explain your results.

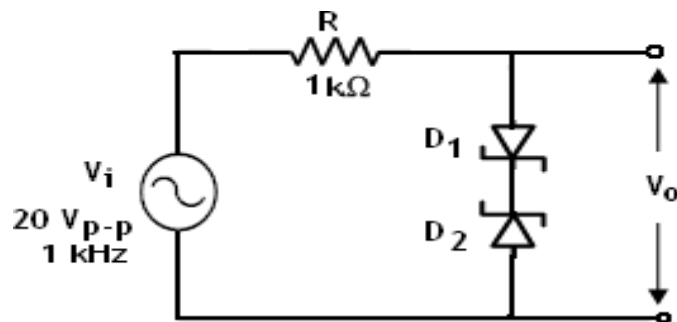
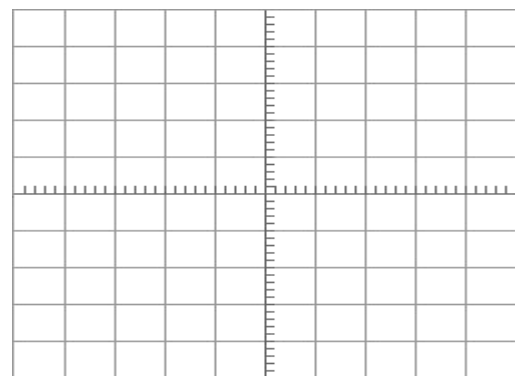
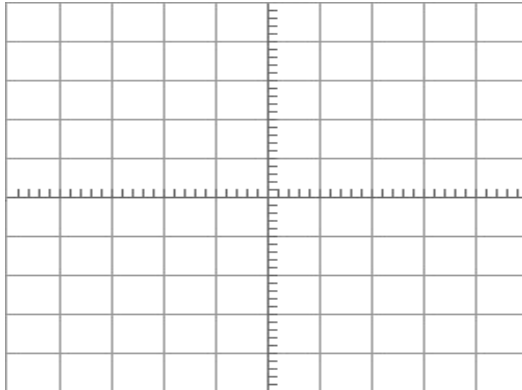


Figure (d): Zener Diode Limiter

8. Connect the circuit as shown in Figure (d).
9. Apply sine wave of 20Vp-p, 1 kHz from function generator.
10. Observe the input, output waveforms and its transfer characteristics on CRO screen.

Volt/div =

Time/div =



Graph 1: Output waveform for circuit in fig (d).

Graph 2: Transfer Characteristics (XY mode)

7.5 Tabular Column:

Table 7.1: Zener Characteristics

E_s (V)	V_{OUT} (V)	$I_Z = V_{RS}/R_S$ (mA)
5		
5.5		
6.00		
6.05		
6.10		
6.15		
6.20		
6.25		
6.30		
6.35		
6.40		
6.45		
6.55		
6.60		
7.00		
8.00		
9.0		
10		
11		
12		
13		
14		
15		

Table 7.2: data for Zener as Voltage Regulator

Parameter	Expected value	Measured value	% error
I_s			
I_z			
I_L			
V_o (with Load)			
V_o (No Load)			

- 1- Replace E_s by an Ac input voltage as shown in figure (e).
- 2- Set v_{in} to (10vp/100Hz). Draw the output and explain the results.

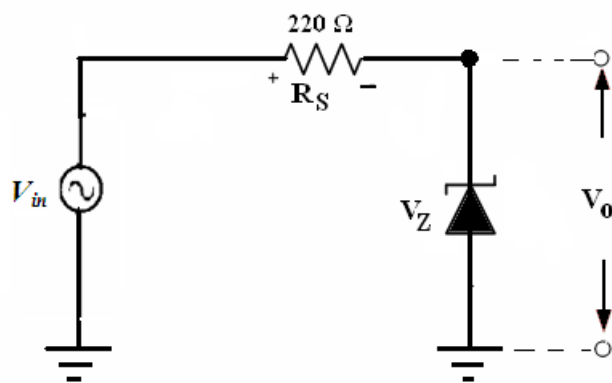


Figure (e): Zener With Ac input

EXPERIMENT EIGHT

BJT OPERATING REGIONS (Transistor as A Switch) & FIXED - BIASED CONFIGURATION

8.1 Objectives:

- To demonstrate the BJT operating regions the saturation, active and the cutoff regions.
- To demonstrate the fixed bias configuration as one of the transistor configuration methods.

8.2 Components & Equipment required:

- Dual DC power Supply 1K Ω , 10K Ω , 100K Ω , 560K Ω
- DMM 2N3904 npn silicon Transistor
- Breadboard Jumpers
- Function Generator
- Dual Trace Oscilloscope.

8.3 Theory:

When operating the transistor, a dc biasing is necessary to establish the proper region of operation for ac amplification. The emitter layer is heavily doped, the base lightly doped, and the collector only lightly doped. The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material. If only one carrier is employed (electron or hole), it is considered a unipolar device.

Transistor Biasing is the process of setting a transistors DC operating voltage or current conditions to the correct level so that any AC input signal can be amplified correctly by the transistor. A transistors steady state of operation depends a great deal on its base current, collector voltage, and collector current and therefore, if a transistor is to operate as a linear amplifier, it must be properly biased to have a suitable operating point. Establishing the correct operating point requires the proper selection of bias resistors and load resistors to provide the appropriate input current and collector voltage conditions. The correct biasing point for a bipolar transistor, either NPN or PNP, generally lies somewhere between the two extremes of operation with respect to it being either “fully-ON” or “fully-OFF” along its load line. This central operating point is called the “Quiescent Operating Point”, or Q-point for short.

When a bipolar transistor is biased so that the Q-point is near the middle of its operating range, that is approximately halfway between cut-off and saturation, it is said to be operating as a Class-A amplifier. This mode of operation allows the output current to increase and decrease around the amplifiers Q-point without distortion as the input signal swings through a complete cycle. In other words, the output current flows for the full 360° of the input cycle.

The fixed-bias circuit of Figure (a) provides a relatively straightforward and simple introduction to transistor dc bias analysis. Even though the network employs an npn

transistor, the equations and calculations apply equally well to a pnp transistor configuration merely by changing all current directions and voltage polarities.

8.4 Procedure:

Part A: The BJT Operating Regions

1. Construct the circuit of Figure 5.1. Adjust V_{BB} to 0V. This will set $I_B = \text{Zero}$,
2. Set V_{CC} to 5V, calculate and then measure I_C and V_{CE} . Record your results in table 8.1.
3. Set I_B to $30\mu\text{A}$, calculate and then measure I_C and V_{CE} .
4. Set I_B to max by setting V_{BB} to 6V, and still maintaining V_{CC} to 5V. Calculate and measure I_C and V_{CE} .

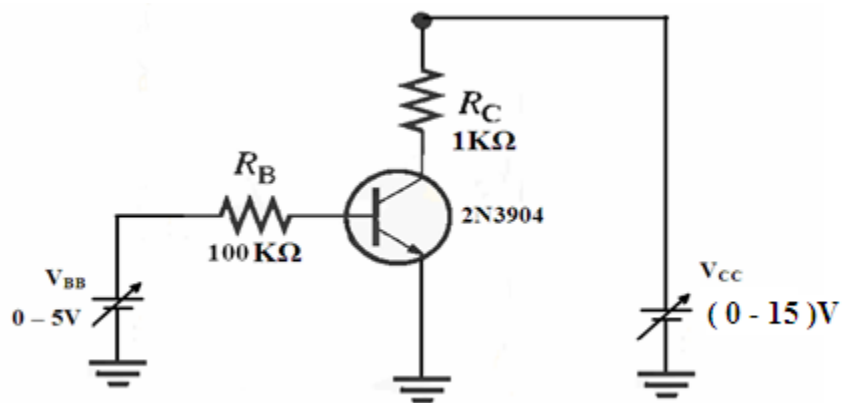


Figure (a): Collector Output Characteristics

5. Replace V_{BB} by $(10\text{Vp} / 100\text{Hz})$ and V_{CC} to 5V as shown in figure (b).
6. Show and draw the output signal. Explain your results.

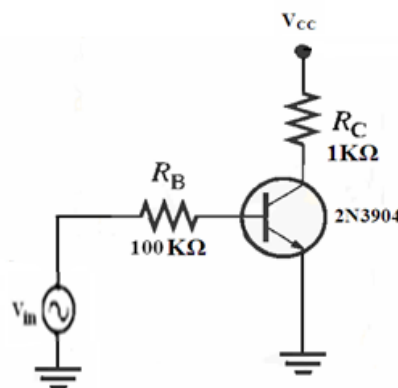


Figure (b): BJT Operating Regions

Part B: Fixed-Biased Configuration

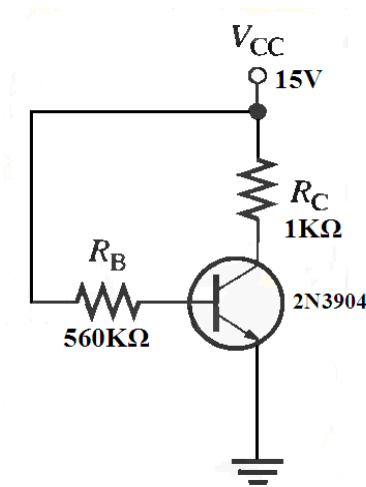


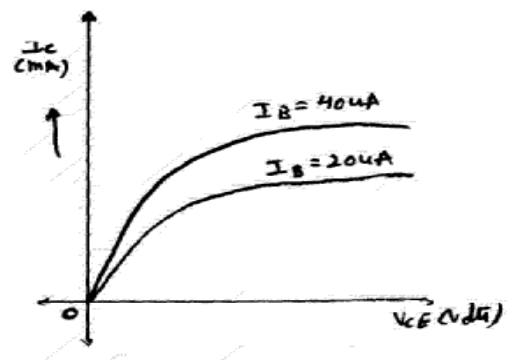
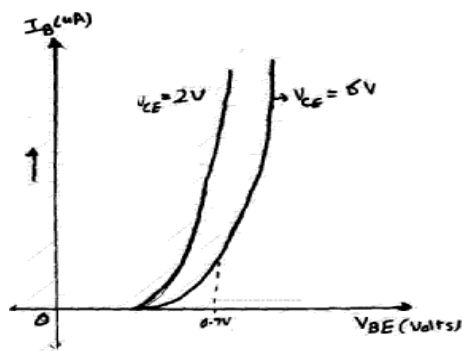
Figure (c): Fixed-Bias Configuration

1. Construct circuit as of Figure 5.3 using 2N3904 transistor and set $V_{CC} = 15\text{ V}$.
2. Measure the voltages V_B and V_{RC} , and then record them in table 5.2.
3. Calculate the resulting base current, I_B and collector current, I_C . Using the values obtained, find β .
4. Using the values obtained in Step 3; calculate the values of V_B , V_C , V_E and V_{CE} .
5. Heat up the transistor in Figure 5.2, and then measure V_B , V_C , V_E and V_{CE} .
6. How do the measured values (Step 5) compare to the calculated values (Step 4)?

8.5 Expected Graphs:

Input Characteristics

Output Characteristics



8.6 Tabular Column:

Table 8.1: Collector Output Characteristics

$I_B (\mu A)$	Calculated		Measured	
	$V_{CE}(V)$	$I_C(mA)$	$V_{CE}(V)$	$I_C(mA)$
0				
30				
max				

Table 8.2: Self-Biased Configuration Results

Parameter	Transistor		Notes
	Expected Value	Measured Value	
I_B			
I_C			
β_{DC}			
V_{BE}	0.7V typical		
V_{CE}			

8.7 PRECAUTIONS

1. Keep the knobs of supply voltages V_{BE} & V_{CE} at minimum positions when switching ON or switching OFF the power supply.
2. No loose contacts at the junctions.
3. Do not overload the meters above its rated ranges.

EXPERIMENT NINE

TRANSISTOR CE CHARACTERISTICS

9.1 Objectives:

- Plot the input and output characteristics of a transistor connected in Common Emitter configuration.
- Calculate the input resistance R_i at $I_B = 20 \mu A$, output resistance R_o at $V_{CE} = 10V$ and current gain at $V_{CE} = 10V$.

9.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1.	Transistor CE trainer Board Containing: a) DC Power Supply. b) PNP Transistor c) Carbon Film Resistor.	(0-12) V BC 107 470 Ω , 1/2 W 100K Ω , 1/2W	2 1 1
2.	a) DC Voltmeter b) DC Voltmeter	(0-1) V (0-20) V	1 1
3.	DC Ammeter	(0-50) mA (0-200) μA	1 1
4.	Connecting wires	5A	12

9.3 Theory:

A transistor is a three terminal device. The terminals are emitter, base, collector. In common emitter configuration, input voltage is applied between base and emitter terminals and output is taken across the collector and emitter terminals. Therefore the emitter terminal is common to both input and output.

The input characteristics resemble that of a forward biased diode curve. This is expected since the Base-Emitter junction of the transistor is forward biased. As compared to CB arrangement I_B increases less rapidly with V_{BE} . Therefore input resistance of CE circuit is higher than that of CB circuit.

The output characteristics are drawn between I_C and V_{CE} at constant I_B . the collector current varies with V_{CE} upto few volts only. After this the collector current becomes almost constant, and independent of V_{CE} . The value of V_{CE} up to which the collector current changes with V_{CE} is known as Knee voltage. The transistor always operated in the region above Knee voltage, I_C is always constant and is approximately equal to I_B .

The current amplification factor of CE configuration is given by

$$B = \Delta I_C / \Delta I_B$$

9.4 Procedure:

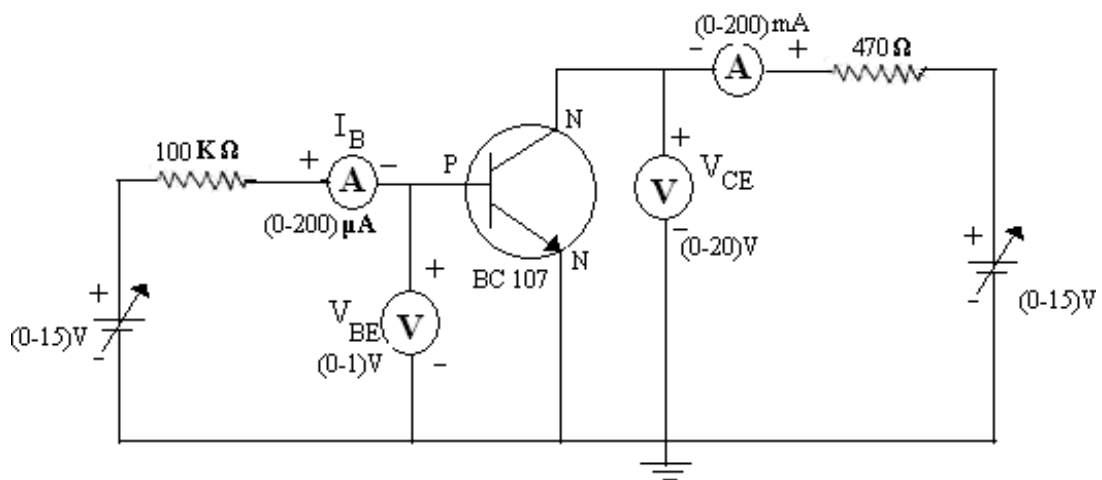
Input Characteristics:

1. Connect the transistor as shown in figure.
2. Keep the V_{CE} constant at 2V and 6V.
3. Vary the I_B in steps and note down the corresponding V_{BE} values as per tabular column.

Output Characteristics:

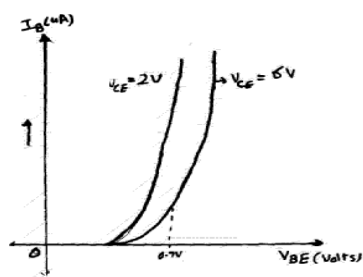
1. Keep the I_B constant at 20 μA and 40 μA .
2. Vary the V_{CE} in steps and note corresponding I_C values.
3. Readings are tabulated as shown in tabular column.

9.5 Circuit Diagram:

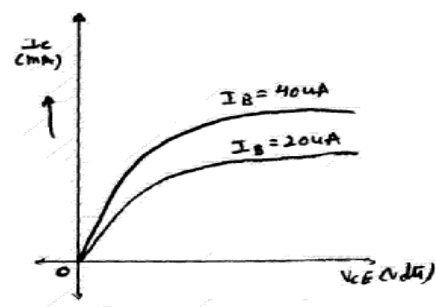


9.6 Expected Graphs:

Input Characteristics



Output Characteristics



9.7 Tabular Column:

[illegible][illegible]

9.8 Precautions:

1. Keep the knobs of supply voltages V_{BE} & V_{CE} at minimum positions when switching ON or switching OFF the power supply.
2. No loose contacts at the junctions.
3. Do not overload the meters above its rated ranges.

9.9 Calculations:

Input Resistance ($I_B=20\mu A$) = $\Delta V_{BE}/\Delta I_B =$

At $V_{CE} = 2V$

Input Resistance ($I_B = 20\mu A$) = $\Delta V_{BE} / \Delta I_B =$

At $V_{CE} = 6V$

Output resistance ($V_{CE}=10V$) = $\Delta V_{CE}/\Delta I_C =$

At $I_B = 20 \mu A$

Output resistance ($V_{CE}=10V$) = $\Delta V_{CE}/\Delta I_C =$

At $I_B = 20 \mu A$

Current Amplification Factor " β " = $\Delta I_C / \Delta I_B =$

H-PARAMETER CALCULATIONS

$$h_{ie} = \Delta V_{be} / \Delta I_b =$$

$$h_{oe} = \Delta I_c / \Delta V_{ce} =$$

$$h_{fe} = \Delta I_c / \Delta I_b =$$

$$h_{re} = \Delta V_{be} / \Delta V_{ce}$$

9.10 Questions:

1. What is the range of β for the transistor?
2. What are the input and output impedances of CE configuration?
3. Identify various regions in the output characteristics?
4. What is the relation between α and β ?
5. Define current gain in CE configuration?
6. Why CE configuration is preferred for amplification?
7. What is the phase relation between input and output?
8. Draw diagram of CE configuration for PNP transistor?
9. What is the power gain of CE configuration?
10. What are the applications of CE configuration?

9.11 Lab Assignment:

Plot the I/O characteristics of CE configuration for $V_{CC} = 10V$, $V_{BB} = 4V$, $R_b = 200K\Omega$, $R_c = 2K\Omega$, $\beta = 200$, $V_{be} = 0.7V$.

9.12 Result:

1. Input and Output curves are plotted.
2. R_i , Input Resistance:
 - a. $V_{CE} = 2V$ and $I_B = 20\mu A$, $R_i =$
 - b. $V_{CE} = 6V$ and $I_B = 20\mu A$, $R_i =$
3. R_o , Output Resistance:
 - a. $V_{CE} = 10V$ and $I_B = 20\mu A$, $R_o =$
 - b. $V_{CE} = 10V$ and $I_B = 40\mu A$, $R_o =$
4. Current Amplification factor
" β " = (at $V_{CE} = 10V$)

EXPERIMENT TEN

TRANSISTOR CB CHARACTERISTICS

10.1 Objectives:

Plot the input and output characteristics of a transistor connected in Common Base configuration. Calculate the input resistance R_i at $I_E = 12\text{mA}$, output resistance R_o at $V_{CB} = 8\text{V}$ and current gain at $V_{CB} = 6\text{V}$.

10.2 Components & Equipment required:

#	Device	Range / Rating	Quantity (in No.s)
1.	Transistor CB trainer Board Containing: a. DC Power Supply. b. PNP Transistor c. Carbon Film Resistor	(0-12)V CK100 470Ω, 1/2 W	2 1 2
2.	a. DC Voltmeter b. DC Voltmeter	(0-1) V (0-20) V	1 1
3.	DC Ammeter	(0-50) mA	2
4.	Connecting wires	5A	12

10.3 Theory:

A transistor is a three terminal active device. The terminals are emitter, base, collector. In CB configuration, the base is common to both input (emitter) and output (collector). For normal operation, the E-B junction is forward biased and C-B junction is reverse biased. In CB configuration, I_E is +ve, I_C is -ve and I_B is -ve.

So, $V_{EB} = f_1(V_{CB}, I_E)$ and $I_C = f_2(V_{CB}, I_E)$

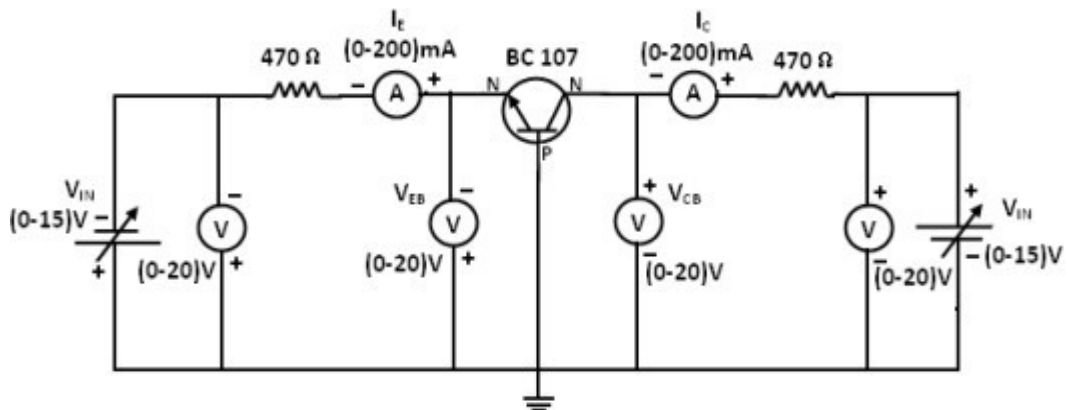
With an increasing the reverse collector voltage, the space-charge width at the output junction increases and the effective base width „W“ decreases. This phenomenon is known as “Early effect”. Then, there will be less chance for recombination within the base region. With increase of charge gradient within the base region, the current of minority carriers injected across the emitter junction increases. The current amplification factor of CB configuration is given by, $\alpha = \Delta I_C / \Delta I_E$.

10.4 Procedure:

- Input Characteristics:
 1. Connect the transistor as shown in figure.
 2. Keep the V_{CB} constant at 4V and 8V. Vary the V_{EB} in steps and note corresponding I_E values as per tabular form.
- Output Characteristics:
 1. Keep the I_E constant at 4mA and 8mA. Vary the V_{CB} in steps and note corresponding I_C values.

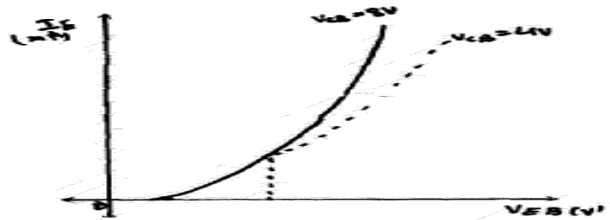
2. Readings are tabulated as shown in tabular column.

10.5 Circuit Diagrams:

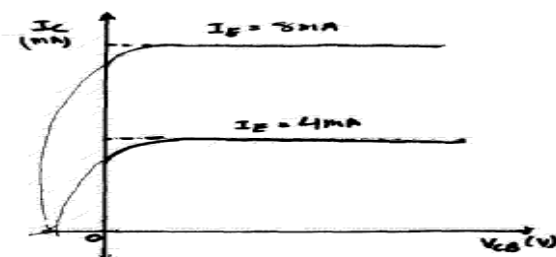


10.6 Expected Graphs:

- Input Characteristics:



- Output characteristics:



10.7 Tabular Column:

Input Characteristics				Output Characteristics			
$V_{CB} = -4V$		$V_{CB} = -8V$		$I_E = 8mA$		$I_E = 4mA$	
V_{EB} (Volts)	I_E (mA)	V_{EB} (Volts)	I_E (mA)	V_{CB} (Volts)	I_C (mA)	V_{CB} (Volts)	I_C (mA)

10.8 Precautions:

1. Keep the knobs of supply voltages V_{EB} & V_{CB} at minimum positions when switching ON or switching OFF the power supply.
2. No loose contacts at the junctions.
3. Do not overload the meters above its rated ranges.

10.9 Calculations:

Input Resistance ($I_E = 12 \text{ mA}$) = $\Delta V_{EB} / \Delta I_E$ At $V_{EB} = 4 \text{ V}$

Input Resistance ($I_E = 12 \text{ mA}$) = $\Delta V_{EB} / \Delta I_E$ At $V_{EB} = 8 \text{ V}$

Output resistance ($I_E = 8 \text{ mA}$) = $\Delta V_{CB} / \Delta I_C$ At $V_{CB} = -8 \text{ V}$.

Output resistance ($I_E = 4 \text{ mA}$) = $\Delta V_{CB} / \Delta I_C$ At $V_{CB} = -8 \text{ V}$.

Current Amplification Factor „ α “ = $\Delta I_C / \Delta I_E$ =

▪ H-parameter calculations

$$h_{ib} = \Delta V_{eb} / \Delta I_e =$$

$$h_{ob} = \Delta I_c / \Delta V_{cb} =$$

$$h_{fb} = \Delta I_c / \Delta I_e =$$

$$h_{rb} = \Delta V_{eb} / \Delta V_{cb} =$$

10.10 Questions:

1. What is the range of α for the transistor?
2. Draw the input and output characteristics of the transistor in CB configuration?
3. Identify various regions in output characteristics?
4. What is the relation between α and β ?
5. What are the applications of CB configuration?
6. What are the input and output impedances of CB configuration?
7. Define α (alpha)?
8. What is EARLY effect?
9. What is the power gain of CB configuration

10.11 Lab Assignment:

Plot the I/O characteristics of CB configuration for $V_{cc} = 12 \text{ V}$, $V_{EE} = 6 \text{ V}$,
 $R_E = 100 \text{ K ohms}$, $R_c = 1 \text{ K ohms}$, $\alpha = 0.98$, $V_{be} = 0.7 \text{ V}$.

10.12 Result:

Input and output curves are plotted.

1. R_i Input Resistance:

i. $V_{EB} = 4 \text{ V}$ and $I_E = 12 \text{ mA}$, $R_i =$

ii. $V_{EB} = 8 \text{ V}$ and $I_E = 12 \text{ mA}$, $R_i =$

2. R_o Output Resistance:

i. $V_{CB} = 8 \text{ V}$ and $I_E = 8 \text{ mA}$, $R_o =$

ii. $V_{CB} = 8 \text{ V}$ and $I_E = 4 \text{ mA}$, $R_o =$

3. Current Amplification factor

“ α ” =

(at $V_{CB} = 6 \text{ V}$)

EXPERIMENT ELEVEN

FET CHARACTERISTICS

11.1 Objectives:

- Find drain resistance at $V_{DS} = 3V$ for $V_{GS} = 0.5V, 0V, -1V$.
- Find Trans conductance at $V_{GS} = -1.0V$ for $V_{DS} = 1V, 2V, 3V$.

11.2 Components & Equipment required:

#	Device	Range / Rating	Quantity (in No.s)
1	FET Characteristics trainer board containing a) DC power supply b) DC power supply c) FET d) Carbon film resistor e) Carbon film resistor	0-12V 0-5V BFW 11 1K Ω , 1/2W 470K Ω , 1/2W	1 1 1 1 1
2	DC Voltmeter	0-20V	2
3	(a) DC Ammeter (b) DC Ammeter	0-200 μ A 0-20 μ A	1 1
4	Connecting wires		12

11.3 Theory:

A FET is a three terminal device, having the characteristics of high input impedance and less noise, the Gate to Source junction of the FET is always reverse biased. In response to small applied voltage from drain to source, the n-type bar acts as sample resistor, and the drain current increases linearly with V_{DS} . With increase in I_D the ohmic voltage drop between the source and the channel region reverse biases the junction and the conducting position of the channel begins to remain constant. The V_{DS} at this instant is called “pinch of voltage”.

If the gate to source voltage (V_{GS}) is applied in the direction to provide additional reverse bias, the pinch off voltage will be decreased.

In amplifier application, the FET is always used in the region beyond the pinch-off.

$$I_{DS} = I_{DSS}(1 - V_{GS}/V_P)^2$$

11.4 Procedure:

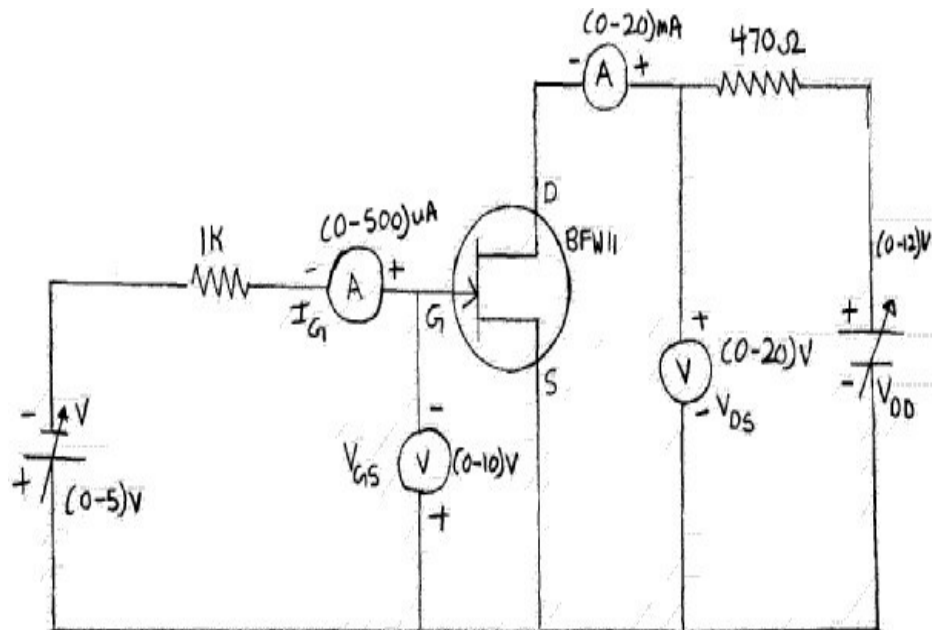
Drain characteristics:

1. Connect the circuit as shown in figure (1)
2. Set the gate source voltage, V_{GS} in 0V position.
3. Increase drain source voltage, V_{DS} in steps and note corresponding I_D values as Shown in the tabular column.
4. Now change V_{GS} to +0.5V and -1V and repeat the above procedure.

Transfer characteristics:

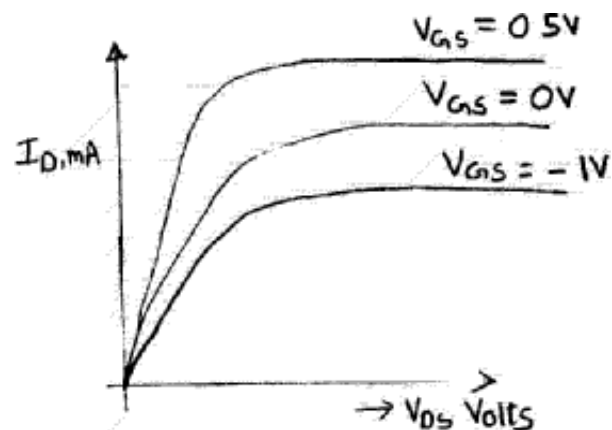
1. Connect the circuit as shown in figure (1).
2. Keep the drain source voltage, V_{DS} at 1V.
3. Vary the gate source voltage, V_{GS} in steps and note corresponding I_D values as shown in the tabular column.
4. Now change V_{DS} to 2V and 3V and repeat the above procedure.

11.5 Circuit Diagram:

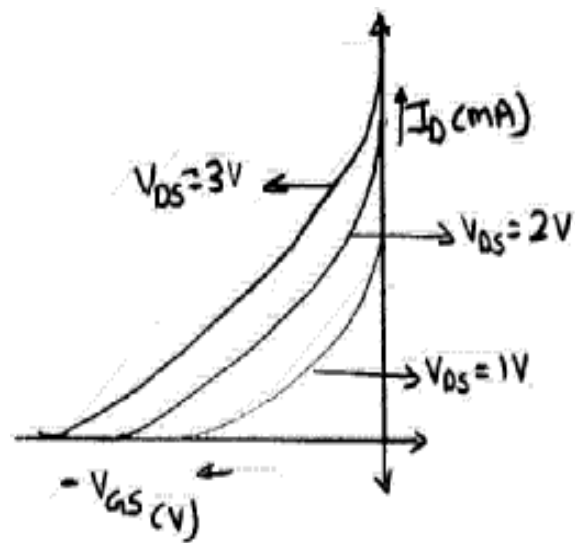


11.6 Expected Graphs:

- Drain Characteristics:



- Transfer Characteristics:



11.7 Tabular Column:

- Drain characteristics:

	$V_{GS}=0V$	$V_{GS}=0.5V$	$V_{GS}=1.0V$
$V_{DS} (V)$	$I_D (mA)$	$I_D (mA)$	$I_D (mA)$
0			
0.2			
0.4			
0.6			
0.8			
1.0			
2.0			
3.0			
4.0			
5.0			
6.0			
7.0			
8.0			
9.0			
10.0			

▪ Transfer characteristics:

	V _{DS} =0V	V _{DS} =2V	V _{DS} = 3V
V _{GS} (v)	I _D (mA)	I _D (mA)	I _D (mA)
0			
-0.2			
-0.4			
-0.6			
-0.8			
-1.0			
-1.2			
-1.4			
-1.6			
-1.8			
-2.0			
-2.2			
-2.4			
-2.6			
-2.8			

11.8 Precautions:

1. Knobs of the supply must be kept at minimum position while switching on and off the supply.
2. Loose connections at junction must be avoided.
3. Device and meter of correct rating and range must be used.

11.9 Calculations:

- AC drain resistance at V_{DS}=3V of V_{GS}= 0.5V
= $\Delta V_{DS} / \Delta I_D$ ii. =
- AC drain resistance at V_{DS}=3V of V_{GS}= 0V
= $\Delta V_{DS} / \Delta I_D$ ii. =
- AC drain resistance at V_{DS}=3V of V_{GS}= -1V
= $\Delta V_{DS} / \Delta I_D$ ii. =
- Transconductance at V_{GS} = -1V of V_{DS}=1V
= $\Delta I_D / \Delta V_{GS}$ ii. =
- Transconductance at V_{GS} = -1V of V_{DS}=2V
= $\Delta I_D / \Delta V_{GS}$
- Transconductance at V_{GS} = -1V of V_{DS}=3V

$$= \Delta I_D / \Delta V_{GS}$$

11.10 Questions:

1. What are the advantages of FET over transistor?
2. Is FET a current controlled device? Explain?
3. What is the operation of a N-channel JFET?
4. Can you compare JFET and a MOSFET?

11.11 Result:

1. At $V_{DS}=3V$
 - Drain Resistance for $V_{GS}=0.5V$ is =
 - Drain Resistance for $V_{GS}=0V$ is =
 - Drain Resistance for $V_{GS}=0.5V$ is =
2. At $V_{GS}=-1V$
 - Transconductance for $V_{DS} = 1V$ is =
 - Transconductance for $V_{DS}= 2V$ is =
 - Transconductance for $V_{DS}= 3V$ is =

EXPERIMENT TWELVE

THE JFET TRANSFER CHARACTERISTIC CURVE

12.1 Objectives:

To use the oscilloscope to display the transfer characteristic curve for the MPF102 JFET. This curve shows the parabolic, or square-law, variation of the drain current as a function of the gate-to-source voltage. From such a curve it is possible to estimate JFET parameters such as the drain current with gate shorted to source, I_{DSS} , the gate-to-source cutoff voltage, $V_{GS(off)}$; and the forward transconductance, g_{m0} .

12.2 Components & Equipment required:

- Resistors (1/4 W): 100Ω.
- MPF102 n-channel JFET
- 1N914 silicon rectifier diode
- 0-15V dc power supply
- Signal generator
- Dual trace oscilloscope
- Breadboarding socket

12.3 Theory:

This experiment demonstrated the shape of the transfer characteristic, or transconductance curve of an MPF102 JFET. Using an oscilloscope as a simple curve tracer, the transfer characteristic curve resembled a parabola showing the nonlinear variation of the JFETs drain current as a function of gate-to-source voltage. From this curve, it was then possible to estimate JFET parameters such as g_{m0} , I_{DSS} and $V_{GS(off)}$, which would be needed for several later experiments.

12.4 Useful Formula:

JFET forward transconductance at $V_{GS} = 0$

$$g_{m0} = \frac{2I_{DSS}}{|V_{GS(off)}|}$$

12.5 Procedure:

1. Wire the circuit shown in the schematic diagram of Figure 12-1. In this part, the oscilloscope is set up to function as an X-Y plotter. Set the oscilloscope's controls to the following approximate settings:
 - Vertical (or Y) input sensitivity: 0.2 V/division, dc coupling
 - Horizontal (or X) input sensitivity: 1 V/division, dc coupling
2. After the oscilloscope has warmed up, move the trace dot to the upper right-hand corner of the oscilloscope's screen so that it is centered on the intersecting scale divisions of the display. Now apply power and the signal generator to the breadboard.

- Adjust the frequency of the signal generator to 500Hz and at a signal level sufficient to produce a display similar to that shown in Figure 12-3. If you do not obtain this curve, the oscilloscope leads may be interchanged or there may be wiring error.

The “horizontal input” measures the JFETs instantaneous gate-to-source voltage (V_{GS}). The diode allows only negative voltage variations, which serve as the drain-to-source voltage. The “vertical input” measures the voltage drop across the 100- Ω resistor. As shown on the display, the vertical axis increases downward, which is inverted from the normal sense. Using Ohm's law, we can make the vertical input read the JFETs instantaneous drain current (I_{DSS}). If the vertical sensitivity is 0.2V/division, then, in terms of the current through the 100- Ω resistors (which is the same as the drain current).

$$\begin{aligned}\text{Vertical Sensitivity} &= (0.2\text{V/division})/100\Omega \\ &= 2\text{mA/division}\end{aligned}$$

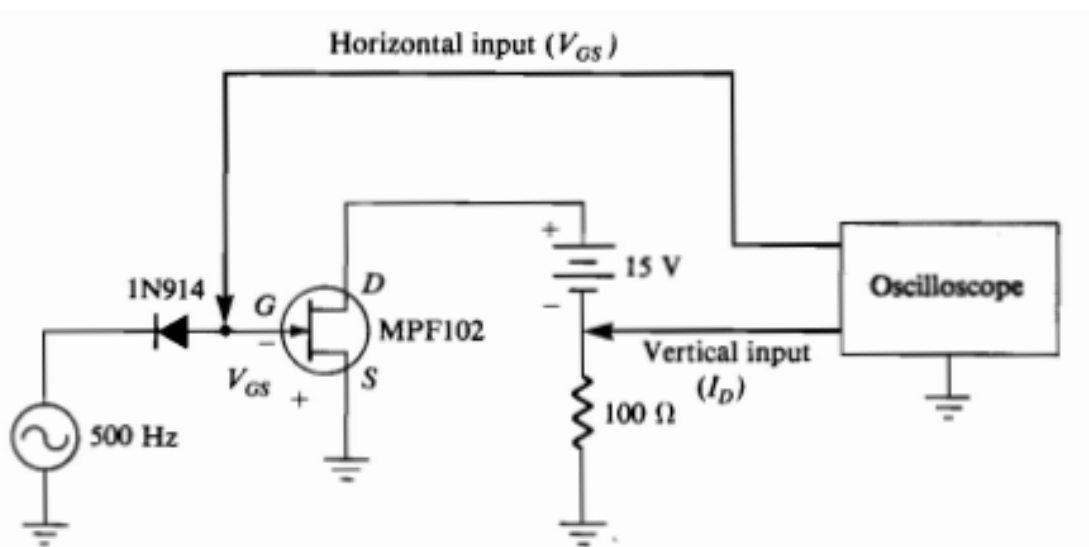


Figure 12-1: Schematic diagram of circuit

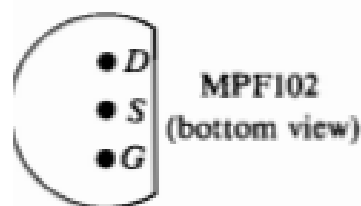


Figure 12-2: Pin configuration of MPF102 JFET

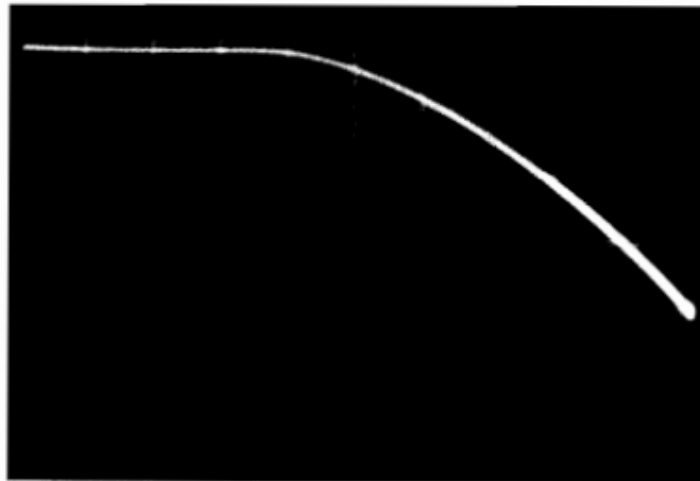


Figure 12-3

4. Using Figure 12-4 as a guide, estimate from the oscilloscope's display both I_{DSS} and $V_{GS(off)}$, for the MPF102 JFET you are using. Record these values in Table 12-1.

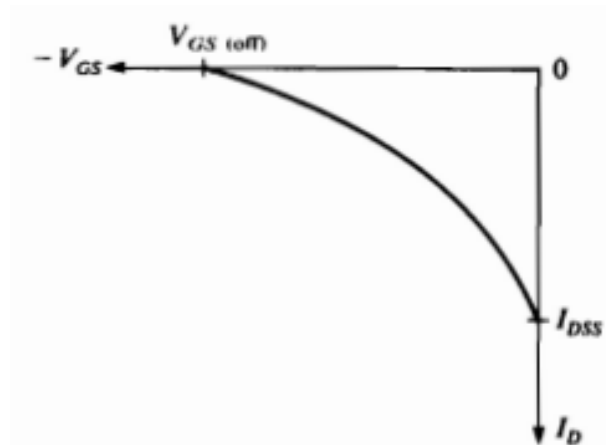


Figure 12-4

5. From the values for I_{DSS} and $V_{GS(off)}$ measured in Step 4, calculate the MPF102 JFET forward transconductance at zero gate- to-source voltage, g_{m0} , and record this value in Table 12-1.

12.6 Tabular Column:

I_{DSS}	
$V_{GS(off)}$	
g_{m0}	

12.7Results:

1. The point at which the curve of Figure 12-3 intersects the vertical axis is:
(a) g_{m0} (b) V_p (c) I_{DSS} (d) $V_{GS(off)}$
2. The point at which the curve of Figure 12-3 intersects the horizontal axis is:
(a) g_{m0} (b) V_{DS} (c) I_{DSS} (d) $V_{GS(off)}$
3. The diode in the test circuit of Figure 12-1 is used to:
(a) Limit the peak drain current
(b) Forward bias the gate-to-source junction
(c) Reverse bias the gate-to-source junction
(d) Protect the JFET from excessive gate voltages
4. From the curve of Figure 12-3, one is able to determine:
(a) I_{DSS} (b) $V_{GS(off)}$ (c) g_{m0} (d) all of the above
5. If $g_{m0} = 5000\mu S$ and $V_{GS(off)} = -3V$, then I_{DSS} is:
(a) 0.6mA (b) 1.2mA (c) 7.5mA (d) 15mA